Shareholder Voice and Executive Compensation*

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Abstract

I estimate a model of CEO compensation with nonbinding shareholder approval votes (Say-on-Pay). Because the compensation proposal is endogenous to the voting environment itself, the threat of dissent disciplines pay ex ante. The estimated disciplinary channel lowers CEO pay by 4.77% and raises shareholder value by 2.22% on average, despite a 6% failure rate. I also analyze a counterfactual vote design that strengthens the communication channel of SOP by letting the information contained in a failed vote directly affect within-period compensation. Relative to the baseline, the design increases shareholder value.

Keywords: shareholder voting, corporate governance, executive compensation, CEO compensation, say-on-pay, structural estimation

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1. Introduction

Shareholders elect the Board of Directors but the Board need not represent their interests (Shleifer and Vishny, 1997). A primary manifestation of this problem is in executive compensation: the Board should set compensation to align the interests of management and shareholders, yet directors may have an incentive to favor the CEO (Bebchuk and Fried, 2003).

When shareholders disagree with compensation decisions and exerting control is not viable, shareholders can convey dissent through *voice* (Hirschman, 1970; Cuñat et al., 2016). Say-on-Pay (SOP), a nonbinding shareholder approval vote on CEO compensation policy, is an important channel through which shareholders may voice dissent. While advisory, SOP functions as a public performance evaluation of the compensation committee and a vote of confidence in the Board's pay-setting choices.

Yet the impact of SOP is unclear. Compensation policies receive over 90% support on average and only about 6% of SOP votes in the US fail.[1] This support may be difficult to reconcile with (i) the literature suggesting that CEOs may influence the pay-setting process (e.g., Bertrand and Mullainathan, 2001; Coles et al., 2014) and (ii) recent survey evidence showing that institutional investors believe CEO pay levels may be too high (Edmans et al., 2023).[2]

An important consideration is that these approval votes are ex post, endogenous outcomes, occurring after compensation has been set and firm performance realized. The pay package brought to a vote is itself an equilibrium object: anticipating how shareholders translate pay and realized performance into dissent, and internalizing the cost of that dissent, the Board may set compensation to make the vote pass. Put differently, the proposal put forth is endogenous to the preferences and technology embedded in the vote itself, and confirming whether low dissent ex post is consistent with effective governance ex ante is an empirical question.

[1] SOP was formalized in the US as part of Dodd-Frank in 2010. SOP proposals are put forth by management at the annual shareholder meeting, and shareholders vote on the CEO's compensation from the previous fiscal year. I use "failure" to refer to SOP proposals that do not garner the required support from shareholders. SOP votes are nonbinding: there is no threshold which forces the Board to change pay policy. However, the understood threshold for SOP failure, by both academics and practitioners, is 70% support (or 30% voting against, see Dey et al., 2024; ISS, 2022, Section 5 "Compensation"). Further, SOP votes are ex post approval votes on the previous year's CEO compensation, not advisory votes on proposed compensation.

[2] Edmans et al. (2023) survey UK institutional investors; one question they ask is "Do you believe the overall level of CEO pay is too low, too high, or about right?" 77% of investors view pay as too high, and 86% of those agreed that "boards are ineffective at lowering it even though they should." See Section 4.4 and Table OA.4 of the paper.

This paper's goal is to pose and estimate a structural model to quantify the influence of SOP on compensation policy and explore the mechanism by which this influence occurs. In the model, the Board decides on compensation policy and may differ from shareholders on the appropriate pay level for the CEO. Shareholders hold the vote: failure may punish the Board, yet dissent from the Board on a prominent policy may itself be costly to shareholders. Estimating the model will quantify the disciplinary influence of SOP on compensation policy and, more generally, the disciplinary efficacy of shareholder voting, by identifying how proposals are shaped ex ante by the anticipated threat of dissent and whether and how governance frictions make dissent costly, thereby tempering discipline.

However, other factors also influence compensation and voting decisions. The size (or existence) of the difference in preferences between the Board and shareholders over pay levels is not obvious. Some CEOs are more skilled than others and will thus be better compensated. The Board and shareholders cannot observe CEO skill directly and may have different beliefs about the CEO's ability (Taylor, 2010); public and private information will influence learning, and consequently, pay and vote decisions. Separately quantifying the effects of these forces is necessary to fully understand the impact of the vote.

I estimate the model by indirect inference. Identification comes from matching reduced-form moments that summarize the equilibrium interaction between the Board and shareholders: a failure probit mapping pay and performance into dissent, and a wage-setting equation mapping anticipated dissent into compensation. The estimated model closely matches the data, in particular the SOP failure rate and the sensitivity of failure to pay and firm performance, the primary determinants of SOP outcomes (Fisch et al., 2018).

The estimation produces several results. To start, I estimate the magnitude of the difference in preferences over the appropriate level of CEO pay, which I refer to as the "governance wedge" between the Board and shareholders. The wedge implies that Boards behave as if they maximize a weighted average of shareholder profits and CEO utility, with the CEO's weight growing modestly over tenure. While the magnitude and interpretation of this difference is not the focus of the paper, it implies there is scope for SOP to provide disciplinary value to shareholders.

For the vote to affect compensation policy, however, it must be that the Board internalizes the threat of vote failure into their compensation decisions. To explain observed behavior, I estimate that Boards internalize a utility cost from SOP failure that is equivalent to 1.62% of shareholder value. While the unconditional failure rate of about 6% means the cost is closer to 0.09% of value in expectation, the threat of costly SOP failure disciplines CEO compensation, even if failure may be ex ante unlikely.

Second, failed SOP votes are considered costly by shareholders. I estimate that shareholders in the model internalize a cost to SOP failure equivalent to 1.40% of value (about 0.08% in expectation). This aligns with survey evidence from Edmans et al. (2023): shareholders state that failing the SOP may be undesirable, for example because they are hesitant to dissent from the Board on a prominent policy.

I estimate that providing SOP as a disciplinary mechanism for shareholders lowers pay levels by 4.77% and improves shareholder value by 2.22% on average.[3] These findings are consistent with prominent papers on SOP adoption: Correa and Lel (2016), in a cross-country design on regulatory adoption of SOP, find that the introduction of SOP lowered pay levels by about 7%; and Cuñat et al. (2016), in a close-vote design on US SOP adoption pre-Dodd-Frank, find that it raised firm valuations by about 5%.

These findings identify the mechanism behind the adoption-based evidence. In the model, the compensation policy brought to a vote is an equilibrium object: the Board endogenously selects proposals to manage ex ante failure risk. My results show that infrequent dissent is consistent with effective discipline because discipline operates primarily through the threat of dissent. This logic extends beyond SOP. In many shareholder votes on board- or management-sponsored policies, such as director elections (Aggarwal et al., 2019), major transactions (Becht et al., 2016), or equity issuance (Holderness, 2018), the proposal on the ballot (or whether a proposal even comes to a vote) may be shaped ex ante by anticipated shareholder dissent.

Say-on-Pay affects compensation policy through two channels. First, the vote aggregates shareholder information and communicates it to the Board. Second, the threat of dissent dis-

[3]These magnitudes come from a counterfactual in which I remove the vote from the structural model and re-simulate under the estimated structural parameters (see Section 5.2). As described in the model, compensation affects the CEO's productive input, so changes in pay translate into changes in output, operating income and, ultimately, shareholder value.

ciplines proposal-setting ex ante. In the baseline advisory regime, both channels operate indirectly: a failed vote does not mechanically change the implemented wage, and vote-revealed information affects pay only in future periods.

I use the estimated model to evaluate a counterfactual that gives SOP a revise-and-resubmit feature, thus allowing vote-contained information to impact compensation policy within-period. In the baseline, a failed vote can affect future pay, but it does not change the compensation already implemented. In the counterfactual, failure triggers a within-period revision: the Board incorporates the information revealed by shareholder dissent and resets pay before implementation. Relative to the baseline, this design lowers average CEO pay by 3.73% and raises shareholder value by 1.28%. The exercise is related to models of shareholder communication with management (e.g., Kakhbod et al., 2023), but here the vote itself is the communication device and dissent informs policy by triggering a within-period policy revision.

While the model captures key institutional features of SOP as a public channel of shareholder voice, it abstracts from two aspects of the setting. First, I do not model shareholder engagement: private communication before or after the annual meeting (e.g., Dey et al., 2024; Heath et al., 2026). Accordingly, my estimates and counterfactuals hold the engagement environment fixed; in practice, engagement would affect outcomes by changing the costs and consequences of public dissent and the degree of information asymmetry between the Board and shareholders. Second, I abstract from strategic interactions among heterogeneous shareholders, which Barry et al. (2026) studies in an empirical model of strategic shareholder voting. Rather than solve a dynamic voting game with pivotality or coordination, I summarize the shareholder base by a representative voting rule that maps observables into a failure probability. This isolates the paper's main mechanism: how boards internalize the threat of dissent when designing compensation proposals.

The paper is organized as follows. I first discuss the paper's contribution to the literature. Section 2 discusses the institutional features of SOP and the data. Section 3 presents the model. Section 4 describes the estimation methodology, with Section 5 showing the results of the structural estimation. Section 6 introduces the counterfactual vote mechanism and discusses its implication for compensation policy and shareholder value. Section 7 concludes.

Literature review. A large empirical literature studies the introduction and consequences of SOP and related pay-vote regulations (e.g., Cai and Walkling, 2011; Ferri and Maber, 2013; Cuñat et al., 2016; Correa and Lel, 2016; Iliev et al., 2015). A central theme is that expanding shareholder voice can reduce pay growth and improve value, but also that realized SOP failures are rare and support rates are high, which complicates inference about the magnitude and channels of SOP's effects (e.g., Armstrong et al., 2013; Kaplan, 2013). My contribution is a structural mechanism: SOP outcomes are endogenous equilibrium objects. The compensation package brought to a vote is chosen anticipating how shareholders translate pay and performance into dissent, so high observed support can be consistent with substantial discipline operating through ex ante incentives rather than frequent ex post failures.

Wagner and Wenk (2025) study policy changes that strengthened binding SOP in Switzerland and document shareholder-value effects across designs. My counterfactual is related: rather than giving shareholders a direct veto, it isolates a communication mechanism in which failed advisory approval forces shareholder information into a revised pay decision. The model therefore provides a mechanism-based, structural account of how strengthening the information channel of advisory voting changes pay-setting and shareholder value.

There is a large literature studying how corporate governance affects firm value. Cuñat et al. (2012) and Flammer (2015) show a causal (positive) relation between adoption of governance-improving mechanisms and firm value; my paper shows how (and by how much) a particular governance mechanism improves shareholder value. Johnson and Swem (2021); Fos (2017) show that, although proxy contests are rare, the threat of initiation influences firm behavior beneficially for shareholders; I show that SOP operates similarly. Holderness (2018) shows that a similar mechanism exists when shareholders vote on equity issuance decisions. Within the realm of hedge fund activism, Gantchev et al. (2019) show that spillover threats of activism induce changes in corporate policies at peer firms; Zhu (2021) focuses on how (threats of) activism influence CEO compensation structures.

This paper relates to the broader literature on shareholder voice (e.g., Hirschman, 1970; Gillan and Starks, 2007), and to work that studies voting as a communication technology whose effectiveness depends on the decision maker's incentives to respond (Levit and Malenko, 2011; Levit, 2019). My results show that recurring votes can matter through equilibrium proposal selection: policies and agendas are shaped ex ante by the anticipated threat of dissent. This logic is not unique to SOP, and connects to evidence that shareholder opposition and governance threats can discipline board- or management-sponsored policies (e.g., director elections, proxy contests, equity issuance, and major transactions, Aggarwal et al., 2019; Fos and Tsoutsoura, 2014; Holderness, 2018; Becht et al., 2016, respectively).

The paper also contributes to structural models of executive compensation, learning, and turnover (e.g., Taylor, 2010, 2013; Page, 2018). I embed a recurring shareholder vote into the pay-setting problem and estimate a tenure-varying governance wedge between boards and shareholders. Whether this wedge allows shareholder voice to create value depends on the equilibrium consequences of dissent, which the estimation quantifies.

A growing structural literature studies shareholder voting. Blonien et al. (2025) study how (the quality) of proxy advisor advice and find that ISS may lead to errors in shareholder voting. Barry et al. (2026) estimate a model of incomplete information to separate blockholder private information and strategic voting considerations from underlying preferences towards management. My contribution is to estimate a dynamic model in which voting and policy choice are jointly determined and in which both sides have private information, and then study the effectiveness of voting mechanisms on influencing policies.

2. Institutional Background and Data

2.1. Institutional Details of Say-on-Pay

SOP became a standard feature of US corporate governance after the 2010 Dodd-Frank Act mandated that public firms provide shareholders with an advisory vote on executive compensation beginning in 2011, with some smaller firms phased in later. In practice, nearly all large US firms hold SOP votes annually. The vote is ex post and advisory: shareholders vote on compensation outcomes disclosed in the proxy statement for the prior fiscal year, rather than on compensation policy going forward. The vote covers executive compensation disclosed under Item 402 of Regulation S-K, including the Compensation Discussion and Analysis (CD&A), which contextualizes the Board's rationale for its compensation policy. The model's action and information-revelation sequence is chosen to replicate these institutional features.

SOP is advisory, so no threshold forces the compensation committee to revise pay. Instead, what matters is the level of dissent at which key governance intermediaries and boards treat the outcome as negative. A central benchmark is 70% support, or 30% votes against. Institutional Shareholder Services (ISS) treats SOP support below 70% as a low-support outcome that triggers heightened scrutiny of the firm's response and may lead ISS to recommend votes against the subsequent SOP proposal and/or directors (ISS, 2022, Section 5, "Compensation"). Consistent with this, firms experience sharp changes in institutional investor engagement around the 70% cutoff, while placebo threshold analyses do not show comparable discontinuities at alternative cutoffs such as 50% support (Dey et al., 2024). For these reasons, throughout the paper I define "SOP failure" as at least 30% of votes cast against the SOP proposal.[4]

2.2. Data and Summary Statistics

For the estimation described in Section 4, I merge vote outcomes from Institutional Shareholder Services (ISS) with CEO compensation from Execucomp and firm-level accounting outcomes from Compustat. The sample covers S&P 1500 firms over 2011–2024. Table 1 reports summary statistics for the key vote, CEO, and firm variables used to construct the targeted moments. SOP votes exhibit high average support: the mean vote against is 8.2%, and 6.1% of votes exceed 30% against, which is the baseline failure rate used as a moment in estimation. CEO total expected pay (Execucomp, TDC1), the measure of pay used in the estimation, averages \$7.47 million, with sizable within-spell changes (interquartile range for annual pay changes of roughly −10% to +18%). The median length of CEO tenure is seven years, which informs the learning processes in the structural model.

Table 1 about here

[4]In untabulated analyses, I show that the likelihood of compensation committee director turnover increases significantly in the year after a SOP vote receiving less than 70% support, with no increase at the 50% margin. This suggests that, for the Board, the 70% margin is more salient.

3. Model

3.1. Technology and Environment

Time is annual. The firm is infinitely-lived and consists of three actors: the Board of Directors, which sets the CEO's pay each period; the CEO, who provides a productive input to the firm in return for pay; and a shareholder base, which holds an approval vote over the Board's pay policy. The wage in the model represents the level of total (expected) pay, with no distinction between base and incentive pay; one interpretation is that the Board and CEO negotiate a pay level that determines the CEO input used in production that year (Taylor, 2013).[5]

To link CEO compensation to firm performance in the simplest possible way, I assume a reduced-form relation between pay and observed output:

$$y_t = \eta A_t w_t^{\alpha}, \tag{1}$$

where w_t is total CEO compensation, $A_t > 0$ is firm productivity, $\alpha \in (0, 1)$ governs the curvature of output in pay, and η maps model output into observed firm revenues. Although pay enters directly in (1), this should be interpreted as a sufficient-statistic representation of an underlying incentive-contract environment. In Internet Appendix IA1.1, I show that if the Board offers a linear contract and the CEO chooses costly input, equilibrium CEO input can be written as a function of total pay. Combining this input-supply relation with a concave production technology yields (1). The reduced-form curvature α therefore summarizes both the CEO's input response to pay and diminishing returns in production.

Firm productivity is centered around CEO skill a but influenced by a mean-zero shock ε_{yt},

$$\ln A_t = a + \varepsilon_{yt}, \quad \varepsilon_{yt} \sim N\left(0, \sigma_y^2\right). \tag{2}$$

Type a is not observed by the Board and shareholders, they make predictions about a based on information they observe (as detailed in Section 3.2). Eq. (2) defines the notion of CEO skill: higher types achieve higher average productivity.

[5]Empirically, w_t is measured by Execucomp's total compensation (TDC1), which bundles salary, bonus, and equity-based pay.

Operating income in year t is

$$\Pi_t = \eta A_t w_t^\alpha - \eta \kappa w_t.$$

The parameter $\kappa > 0$ governs how CEO pay translates into firm-level operating costs.[6] The scale parameter η and the cost parameter κ are not essential for the model's economics, but are important for estimation because they map the model's objects into the revenues and operating income observed in the data. Normalizing operating income by η (so $\pi_t \equiv \Pi_t/\eta$), we have

$$\pi_t = A_t w_t^\alpha - \kappa w_t. \tag{3}$$

The Board of Directors (B) sets the wage w_t. A key friction is that the Board's pay-setting objective may differ from the shareholders' objective. In particular, the Board may place additional weight on CEO compensation relative to shareholder operating income. Abstracting from dynamic considerations and Say-on-Pay, and expressing pay-setting objectives in normalized units, the Board chooses w_t to maximize

$$\pi_t^{\mathrm{B}} = A_t w_t^\alpha - (1 - \lambda_\tau)\kappa w_t,$$

where $\lambda_\tau \in [0, 1)$ captures the Board–shareholder preference wedge. When $\lambda_\tau = 0$, the Board internalizes the full marginal cost of pay. When $\lambda_\tau > 0$, the Board acts as if the marginal cost of pay is lower, and therefore chooses a higher level of compensation than shareholders would prefer. I allow this wedge to vary with CEO tenure according to

$$\lambda_\tau = \lambda_0(1 + \lambda_1)^\tau, \tag{4}$$

with parameters restricted so that the wedge remains in $[0, 1)$ over the observed tenure support. I refer to λ_τ as a governance wedge: it represents a difference between the Board and

[6]When $\kappa = 1$ CEO pay and operating costs move in unison; $\kappa \lessgtr 1$ implies that the cost impact is smaller or larger, respectively.

shareholders over the optimal level of CEO compensation.

The tenure dependence has an institutional interpretation. As CEOs remain in office, they may accumulate influence over director nominations, causing successive appointments to tilt the Board toward CEO-friendly directors. This is the endogenous-director-selection mechanism in Hermalin and Weisbach (1998); consistent with this interpretation, Coles et al. (2014) document that board co-option rises monotonically with CEO tenure. Internet Appendix IA1.1 shows how this mechanism maps into $\lambda_\tau > 0$.[7]

As Taylor (2013) shows, CEO wages display downward rigidity: risk-averse CEOs accept lower total pay if they are protected from downside risk. To match observed patterns in compensation, the model incorporates an adjustment cost into the Board's compensation decision:

$$AC(w_t; w_{t-1}, \tau_t) = c_w \times w_t \left(\frac{w_t - w_{t-1}}{w_t} \right)^2 \times \mathbb{1}[w_t < w_{t-1}] \times \mathbb{1}[\tau_t > 0]. \tag{5}$$

The adjustment cost is quadratic and only activates if the Board decreases the wage from $t-1$ to t, chosen to match downward rigidity in CEO pay; c_w controls the cost of adjustment, and is to be estimated.

Shareholders hold an approval vote each year on the Board's CEO pay policy, or a Say-on-Pay vote. The vote is nonbinding (failure does not mandate a change in compensation policy) and occurs at the end of each period (after the compensation decision and output have occurred, following the observed institutional structure). A failed vote results in a potential cost for the Board, $\chi_b \geq 0$. In the model, this cost measures the Board's (perceived) aversion to a failed vote. The Board's per-period utility is

$$A_t w_t^\alpha - \kappa w_t + \lambda_\tau \kappa w_t - \chi_b \times \mathbb{1}[\text{SOP fail}_t] - AC(w_t; w_{t-1}, \tau_t). \tag{6}$$

Shareholders in the model seek to maximize operating income (3): if the proposed wage is high relative to shareholders' beliefs about CEO ability, shareholders may vote against the pay policy. Upon vote failure, shareholders may also incur a perceived cost, which can capture

[7] Tenure-based pay growth may also reflect learning-by-doing if CEO productivity improves with tenure in ways not fully captured by the model's performance-based updating block. In that case, the model may overstate how much tenure-based pay growth reflects governance frictions rather than skill accumulation.

(for example) reputational or coordination costs of dissenting from the Board on compensation policy. The parameter $\chi_s \geq 0$ governs this cost.[8] Shareholders' per-period utility is

$$A_t w_t^{\alpha} - \kappa w_t - \chi_s \times \mathbb{1}[\text{SOP fail}_t]. \tag{7}$$

Eqs. (6) and (7) summarize the differences between the Board and shareholder objectives. In equilibrium, shareholders can discipline the Board's pay-setting wedge (λ_τ) by threatening to fail the SOP, but this discipline is tempered by the shareholders' own cost of a failed vote. The Board additionally internalizes the adjustment cost.[9]

3.2. Beliefs, Signals and Model Timeline

Each period the firm separates from the CEO with exogenous probability f_τ and matches with a new CEO of tenure $\tau = 0$.[10] At the start of each match, the Board and shareholders begin with the prior about CEO skill,

$$a \sim N\left(\mu_0, \sigma_0^2\right), \tag{8}$$

which matches the distribution of ability in the CEO talent pool. At the annual compensation committee meeting, the Board receives its private signal about the CEO,

$$z_{bt} = a + \varepsilon_{bt}, \quad \varepsilon_{bt} \sim N\left(0, \sigma_{z_b}^2\right). \tag{9}$$

The signal z_{bt} will inform the Board's wage decision, and may represent operational interaction with the CEO. Shareholders do not observe z_{bt}, which means that the Board is asymmetrically informed (more precise beliefs) about CEO skill when setting pay. After pay is set, both output and productivity realize (that is, B does not observe A_t before setting compensation).

At the annual shareholder meeting, dispersed (atomistic) shareholders form assessments

[8] Surveyed institutional investors report reluctance to vote against SOP because they prefer to resolve concerns through engagement rather than public dissent, and because a negative vote can trigger repeated follow-up requests from the firm, "imposing a significant time cost" (Edmans et al., 2023, Online Appendix A).

[9] The assumption that shareholders do not incorporate the adjustment cost helps keep the shareholders' problem static, which greatly simplifies the numerical solution.

[10] The model takes CEO turnover as exogenous and focuses on compensation and voting within a firm-CEO match. This abstraction allows the analysis to isolate how SOP affects pay policies and Board incentives conditional on a given match.

of CEO type using both public information and investor-specific information. These assessments are correlated across shareholders (e.g., through proxy recommendations and common information sources). Rather than modeling an explicit strategic voting game, I summarize the informational content of the shareholder base by a single representative Shareholder, labeled S. Appendix IA1.2 microfounds this reduction by showing that, under correlated information and atomistic ownership, the vote outcome can be represented using a sufficient statistic for the shareholder base's information.

At the end of each period t, S observes an aggregate private signal about CEO type,

$$z_{st} = a + \varepsilon_{st}, \quad \varepsilon_{st} \sim N\left(0, \sigma_{z_s}^2\right). \tag{10}$$

S also observes output y_t and the CEO's pay w_t. These observables imply a public signal about productivity A_t, which is informative of the CEO's type. I label this signal

$$z_{yt} = \ln A_t = a + \varepsilon_{yt}, \quad \varepsilon_{yt} \sim N\left(0, \sigma_y^2\right). \tag{11}$$

Both the private signal z_{st} and the public signal z_{yt} will affect the SOP result.

Model timeline. Figure 1 displays the per-period sequence of events. Within a period, the Board and Shareholder hold different information sets: the Board sees its private signal z_{bt} before setting pay, while the Shareholder sees z_{st}, the public productivity signal z_{yt}, and the wage before voting. By construction, however, beliefs coincide at the start of each period: I assume signal disclosure in the wage-vote environment, so the equilibrium mapping from signals to actions implies that the Board's wage choice and the SOP vote outcome reveal z_{bt} and z_{st}. The shared belief at the start of t is therefore $a \sim N(\mu_{at}, \sigma_{at}^2)$, see Appendix IA1.5. At the compensation committee, B receives its signal z_{bt}, which informs their wage decision. Then, operations take place: the CEO supplies their input, productivity realizes and output occurs. Finally, S receives the private signal z_{st}; output, productivity, and wages are revealed; and the SOP vote occurs.

Figure 1 about here

Board and Shareholder beliefs. Both B and S use Bayes' rule to update beliefs about CEO ability after their signals. I use subscript a to refer to beliefs shared by B and S: $(\mu_{at}, \sigma_{at}^2)$ refers to the shared prior at the beginning t. Subscripts b and s refer to when B and S can have different beliefs. CEO tenure fully determines the variance of beliefs; the function $\sigma_a^2(\tau)$ tracks how the variance of beliefs decreases across tenure. Given tenure τ_t at t,

$$\sigma_{at}^2 = \sigma_a^2(\tau_t) = \sigma_0^2 \left[1 + (\tau_t) \times \sigma_0^2 \left(\sigma_{z_b}^{-2} + \sigma_{z_s}^{-2} + \sigma_y^{-2}\right)\right]^{-1}, \quad \sigma_a^2(0) = \sigma_0^2. \tag{12}$$

The mean evolves according to

$$\mu_{at+1} = \sigma_a^2(\tau_t + 1)\left[\frac{\mu_{at}}{\sigma_a^2(\tau_t)} + \frac{z_{bt}}{\sigma_{z_b}^2} + \frac{z_{st}}{\sigma_{z_s}^2} + \frac{z_{yt}}{\sigma_y^2}\right], \tag{13}$$

and the rate of decline of the variance σ_{at}^2 means that μ_{at} tends toward the CEO's true ability.

3.3. The Say-on-Pay Vote

This section formalizes the disciplining mechanism by detailing the Shareholder's strategy, holding the Board's wage choice fixed. I model the shareholder side as a reduced-form voting technology rather than as a strategic voting game. The economically relevant event for the Board is whether the proposal crosses the institutional support threshold, and the paper's object of interest is how the Board internalizes that threat ex ante. I therefore summarize the shareholder side by a parsimonious mapping from information and proposed pay into a failure probability.

To keep the vote tractable, I make two assumptions. First, I model S' voting strategy as a threshold rule that depends on the information available at the annual meeting and the observed wage; equivalently, S commits to an upper bound on the wage they will approve.[11] Second, S votes only based on contemporaneous pay and signals, rather than solving a dynamic voting problem.[12] This avoids introducing a shareholder value function into the solution,

[11] I interpret commitment to the voting rule as an institutional reduced-form assumption. Large institutional investors and proxy advisors operate through pre-specified voting guidelines and repeated stewardship interactions, making rule-based voting empirically plausible. Appendix IA1.3 provides an illustrative relational-contract rationale, but the main model takes commitment to the voting rule as a primitive.

[12] In untabulated reduced-form results, SOP voting outcomes respond primarily to contemporaneous pay, performance, and revenues, with limited sensitivity to lagged measures.

substantially reducing the computational cost in the estimation.

3.3.1. The Vote Decision

Informally, S fails the SOP when the CEO's wage is "too high" given S' beliefs about CEO type and the cost of vote failure. The notion of "too high" is summarized by a threshold rule: the Shareholder chooses an upper bound on the acceptable wage, and fails if the Board's wage exceeds this bound. Formally, conditional on the wage choice, S' strategy can be represented as the threshold:

$$k_{st} = s_t \times w_t, \tag{14}$$

where s_t is the Shareholder's choice variable. The multiplicative form is the simplest specification that delivers a wage-elastic failure probability, which is the equilibrium object of focus in the estimation. After observing signals at the shareholder meeting, they compute a shareholder-optimal benchmark wage implied by their posterior, which I label w_t^S and define below. The vote fails whenever the benchmark falls below the threshold:

$$\text{SOP fail}_t \qquad \Longleftrightarrow \qquad w_t^S \leq k_{st} = s_t \times w_t.$$

Thus, s_t controls the sensitivity of failure probability to changes in the Board's wage choice w_t: a lower s_t makes shareholders more tolerant of deviations from their benchmark.

At the annual shareholder meeting, the Shareholder receives its private signal z_{st} and the public productivity signal $z_{yt} = \ln A_t$. The Shareholder combines these into a signal about CEO type, which has distribution

$$Z_{syt} = pz_{st} + (1-p)z_{yt} = a + \varepsilon_{syt} \quad Z_{syt} \mid \left(\mu_{at}, \sigma_{at}^2\right) \sim N\left(\mu_{at},\ \sigma_{at}^2 + \sigma_{sy}^2\right), \tag{15}$$

where ε_{syt} precision-weights the private and public shocks and has variance $\sigma_{sy}^2 = \frac{\sigma_{z_s}^2 \sigma_y^2}{\sigma_{z_s}^2 + \sigma_y^2}$.

Distance from the shareholder-optimal benchmark w_t^S. The SOP rule is naturally indexed by the gap between the observed wage and shareholder-optimal wage implied by S' information after the meeting. After seeing meeting-based information Z_{syt} and updating beliefs, the shareholder-optimal wage w_t^S would arise out of the following program:

$$\arg\max_{w_t} \quad \mathbb{E}_{st}\left[\exp\left(a + \varepsilon_{yt}\right)\right] w_t^{\alpha} - \kappa w_t = \left(\frac{\alpha}{\kappa}\mathbb{E}_{st}[\exp\left(a + \varepsilon_{yt}\right)]\right)^{\frac{1}{1-\alpha}}.$$

Ex ante, at the time S chooses its strategy, w_t^S has the following distribution:

$$\begin{aligned} w_t^S \mid (\mu_{at}, \sigma_{at}^2) &\sim \log N(m_t^S, v_t^S), \\ m_t^S &\equiv \frac{\mu_{at} + \log\frac{\alpha}{\kappa} + \frac{1}{2}\left(\sigma_{st}^2 + \sigma_y^2\right)}{1-\alpha}, \qquad v_t^S = \frac{\sigma_{at}^2 - \sigma_{st}^2}{(1-\alpha)^2}. \end{aligned} \tag{16}$$

where $\sigma_{st}^2 = \frac{\sigma_{at}^2\sigma_{sy}^2}{\sigma_{at}^2+\sigma_{sy}^2}$ is the Shareholder's posterior variance. I refer to the CDF of this distribution as F_t^S, and use it to assess the probability of vote failure in the Board and Shareholder's objectives:

$$w_t^S \mid \left(\mu_{at}, \sigma_{at}^2\right) \sim F_t^S.$$

3.3.2. Determining the Probability of Say-on-Pay Failure

Given the threshold rule in (14), the SOP vote fails when the information available at the annual meeting implies that the CEO wage is high relative to S' benchmark wage. Thus, conditional on the Board's wage choice w_t, the ex ante probability of failure is

$$\Pr(\text{SOP fail}_t) = \Pr\left(w_t^S \le s_t \times w_t\right) = F_t^S(s_t \times w_t). \tag{17}$$

As discussed above, SOP is modeled as an ex ante rule that disciplines the Board through the anticipated probability of failure. Accordingly, S chooses the policy parameter s_t ex ante, anticipating the effects that meeting-based information will have on beliefs. Importantly, because the Board observes z_{bt} before choosing w_t, the Shareholder must set their threshold in expectation over the Board's information (i.e., S integrates over the distribution of z_{bt} when choosing s_t), and the Board subsequently chooses $w_t(z_{bt}, s_t)$ given its signal and S' choice. The

Shareholder's problem is therefore

$$\max_{s_t} \quad \int_{z_b} f(z_b) \left[\mathbb{E}_{st}[A_t] w_t(z_{bt}, s_t)^{\alpha} - \kappa w_t(z_{bt}, s_t) - \chi_s \times F_t^S(s_t \times w_t(z_{bt}, s_t)) \right] dz_b. \tag{18}$$

3.4. The Compensation Decision

Each period, the Board observes its private signal z_{bt} and chooses the CEO's wage w_t, taking as given the Shareholder's SOP policy s_t. At the start of t, the common belief about CEO type is $a \sim N(\mu_{at}, \sigma_{at}^2)$. Upon observing z_{bt}, the Board updates its posterior to $\left(\mu_{bt|z_b}, \sigma_{bt|z_b}^2\right)$, where

$$\begin{aligned} \mu_{bt|z_b} &= \sigma_{bt|z_b}^2 \left(\frac{\mu_{at}}{\sigma_{at}^2} + \frac{z_{bt}}{\sigma_{z_b}^2} \right) \\ \sigma_{bt|z_b}^2 &= \left(\sigma_{at}^{-2} + \sigma_{z_b}^{-2} \right)^{-1} = \frac{\sigma_{at}^2 \sigma_{z_b}^2}{\sigma_{at}^2 + \sigma_{z_b}^2}. \end{aligned} \tag{19}$$

The Board's wage policy solves the following Bellman equation:

$$\begin{aligned} V(\mu_{at}, \tau_t, w_{t-1}) = \max_{w_t} \quad & \mathbb{E}_{bt|zb}[A_t] w_t^{\alpha} - (1 - \lambda_\tau) \kappa w_t - \chi_b \times F_t^S(s_t \times w_t) - AC(w_t, w_{t-1}; \tau_t) \\ & + \delta_b \Big[(1 - f_{\tau_t}) \mathbb{E}_{bt|zb}[V(\mu_{at+1}, \tau_t + 1, w_t)] + f_{\tau_t} V^R \Big]. \end{aligned} \tag{20}$$

The state consists of the current shared belief mean μ_{at}, CEO tenure τ_t (which pins down belief dispersion; see Eq. 12), and the previous period's wage w_{t-1} (set to zero when $\tau_t = 0$). The term $F_t^S(s_t \times w_t)$ is the (anticipated) probability that SOP fails, as detailed in Section 3.3. The expectation operator $\mathbb{E}_{bt|z_b}[\cdot]$ is taken with respect to the Board's posterior after observing z_{bt}, i.e., using (19). The hazard function f_{τ_t} governs CEO-firm separation and is taken as an input to the model.[13] Upon separation, the Board hires a new CEO and beliefs reset to (μ_0, σ_0), so the termination value is

$$V^R = V(\mu_0, \tau = 0, 0). \tag{21}$$

[13]Pooling all spells, f_τ is the frequency of turnover after τ years, conditional on surviving $\tau - 1$ years. I follow Taylor (2010) in computing these hazard rates. For tractability, turnover is treated as exogenous; the analysis focuses on pay-setting and SOP discipline within an ongoing CEO-firm match. For simplicity, $f_0 = 0$ in estimation, and $f_\tau = 1$ at the tenure cap $\mathbf{T}$. The estimation sample excludes CEO spells that last only one year.

3.4.1. Shareholder Value

Given the Board's equilibrium wage policy $\omega_t = \omega(\mu_{at}, \tau_t, w_{t-1})$, I evaluate outcomes using a shareholder value criterion: the discounted present value of operating income implied by the equilibrium policy,

$$V^{\mathrm{SH}}(\mu_{at}, \tau_t, w_{t-1}) = \mathbb{E}_{at}[A_t]\omega_t^{\alpha} - \kappa\omega_t + \delta_b\Big[(1 - f_{\tau_t})\mathbb{E}_{at}\big[V^{\mathrm{SH}}(\mu_{at+1}, \tau_t + 1, \omega_t)\big] + f_{\tau_t}V^{\mathrm{SH},R}\Big], \quad (22)$$

with $V^{\mathrm{SH},R} \equiv V^{\mathrm{SH}}(\mu_0, \tau = 0, 0)$. This differs from the Board's objective because it values only operating-income cash flows. The pay-setting wedge λ_τ, the perceived vote and adjustment costs, and SOP enter shareholder value only through their effect on the equilibrium wage policy ω_t. I use (22) to evaluate counterfactual changes to the SOP environment.

3.5. Model Solution

I apply Bayes' rule to characterize the Board's and the Shareholder's posterior beliefs about CEO type and substitute these beliefs into the Board's Bellman equation (20) and the Shareholder's objective (18). I compute a Markov equilibrium in stationary policies as follows. For each Board state (μ_t, τ_t, w_{t-1}), I start from a candidate Shareholder SOP policy parameter s_t. Taking this s_t as given, I solve the Board's dynamic program to obtain the wage policy $\omega(\mu_t, \tau_t, w_{t-1})$. Given the induced wage policy, I then update the Shareholder's best response for s_t. I iterate between these two steps until the wage and vote policies converge. Appendix IA1.6 provides full computational details.

4. Estimation

I estimate the parameters of the structural model using indirect inference (McFadden, 1989): I choose the vector of structural parameters which minimizes the difference between the reduced-form outcomes of an auxiliary model estimated on observed and simulated data. As is standard, I weight moments by the inverse of the estimated variance-covariance matrix of the data moments, clustered by firm–CEO spell. Further details of the estimation are presented in Appendix IA2.

4.1. Identification Strategy

I show there is a tight relation between the reduced-form moments of the auxiliary model and structural parameters. Table A.1 displays the reduced-form outcomes and the parameter(s) each outcome targets.

4.1.1. Output and CEO skill parameters

Because the structural model has neither fixed firm or time heterogeneity, I purge these components from observed revenues using a longer panel (1992–2024) by estimating the auxiliary FE decomposition

$$\log \text{rev}_{fit} = c_{\text{rev}} + \psi_f + \phi_i + \lambda_t + u_{fit},$$

where f, i and t indicate firm, CEO-spell and year, respectively. I then construct the empirical counterpart to the model's definition of log revenues as

$$\log y_{it}^{\text{data}} \equiv \log \text{rev}_{fit} - \hat{\psi}_f - \hat{\lambda}_t.$$

In the structural model,

$$\log y_{it} = \log \eta + a_i + \alpha \log w_{it} + \varepsilon_{yit}.$$

Since $\mathbb{E}[a_i]$ and $\log \eta$ are not separately identified, I normalize $\mu_0 = \mathbb{E}[a_i] = 0$ and target $\mathbb{E}[\log y_{it}^{\text{data}}]$ to pin down $\log \eta$. To discipline $(\alpha, \sigma_y, \sigma_0)$ I use an auxiliary regression on the estimation sample (2011–2024):

$$\log y_{it}^{\text{data}} = y_1 \log w_{it} + \mu_i^y + \epsilon_{it}^y. \tag{23}$$

I target the auxiliary moments $\hat{y}_1$, $\mathrm{Var}(\hat{\epsilon}^y_{it})$, and $\mathrm{Var}(\hat{\mu}^y_i)$.[14] $\hat{y}_1$ is treated as a reduced-form elasticity, and helps identify α. $\mathrm{Var}(\hat{\epsilon}^y_{it})$ identifies σ_y and $\mathrm{Var}(\hat{\mu}^y_i)$ identifies σ_0.

In the model, operating profits are

$$\Pi_{it} = y_{it} - \kappa\eta w_{it}, \qquad y_{it} = \eta A_{it} w^{\alpha}_{it},$$

so the model-implied cost share satisfies the identity $1 - \Pi_{it}/y_{it} = \kappa(\eta w_{it}/y_{it})$. In the data, I define operating costs net of CEO pay:

$$s^c_{it} \equiv \frac{\mathrm{Sale}_{ft} - \mathrm{OIBDP}_{ft} - w_{it}}{\mathrm{Sale}_{ft}}, \qquad s^w_{it} \equiv \frac{w_{it}}{\mathrm{Sale}_{ft}},$$

and estimate the auxiliary regression

$$s^c_{it} - \overline{s^c} = k_1\, s^w_{it} + \epsilon^k_{it}, \tag{24}$$

de-meaning the dependent variable so that only the slope $\hat{k}_1$ is targeted. The structural scale parameter κ is identified from $\hat{k}_1$.

4.1.2. The Board-Shareholder Game

Identifying the parameters relating to the wage and vote decisions is a crucial empirical step, and the identification argument centers on matching equations that describe (in reduced-form) the model's equilibrium interaction between the Board and Shareholder.

The probability of SOP failure. The model's definition of SOP failure is

$$\mathbb{1}[\text{SOP fail}_{it}] = \mathbb{1}\left[s_{it} w_{it} - w^S_t \geq 0\right],$$

where w^S_t is the benchmark wage induced by the Shareholder's posterior post-meeting (see Section 3.3.1), and s_{it} is the Shareholder's choice variable. This implies a probability of failure

[14] To ensure consistency between data and model, I compute these reduced-form objects using the same fixed-effect and residualization procedures in both observed and simulated data, so that any finite-sample estimation noise in μ^y_i and $\hat{y}_1$ is mirrored in the simulated moments.

(conditional on information revealed up to period t, $\mathcal{I}_{it}$) equal to

$$\Pr(\text{SOP fail}_{it} \mid \mathcal{I}_{it}) = \Phi\left(\frac{\log(s_{it}w_{it}) - c_0 - c_1\mu_{it}}{c_1\sigma_{z_s}}\right), \tag{25}$$

where (c_0, c_1) are constants that map (posterior) belief μ_{it} to the shareholder-optimal wage distribution.

First, the unconditional failure rate is informative about the Shareholder's aversion to failure χ_s: a higher χ_s will lead to a lower choice of s_{it} on average and a lower rate of failure across the entire sample. The model-implied determination of SOP failure maps to the probit regression

$$\Pr\left(\text{SOP fail}_{it} \mid w_{it}, \epsilon^y_{it}\right) = \Phi\left(s_0 + s_1 \log w_{it} + s_2\epsilon^y_{it}\right) \tag{26}$$

where ϵ^y_{it} is the residual from the output regression (23).[15]

I target the average marginal effects (AME) implied by s_1 and s_2.[16] The AME of log compensation is highly informative about both χ_s and χ_b: $\hat{s}_1$ is decreasing in χ_s, which dampens the sensitivity of failure to the wage, and increasing in χ_b, as the Shareholder internalizes that SOP failure has a larger impact on compensation (all else equal). As shareholders learn about CEO ability via output innovations, including the output residual ϵ^y_{it} isolates residual variation as attributable to private signal realizations (which I discuss further below).

The Board's compensation decision. The Board's pay choice depends on the Board's beliefs about CEO productivity, the size of the governance wedge, the probability of SOP failure, and an adjustment cost if the Board lowers pay. To start, consider a static version of the Board's problem (i.e., we abstract away from dynamics and the adjustment cost).[17]

[15]Two equilibrium implications motivate the auxiliary probit. First, under Gaussian signals, posterior beliefs are affine in a persistent ability component, output innovations, and the shareholder's private signal. Second, the shareholder's choice s_{it} is an equilibrium object measurable with respect to $\mathcal{I}_{it}$, so $\log(s_{it}w_{it})$ is also an affine function of the same state variables (up to an approximation error). This yields a probit index that is linear in compensation and output innovations, with z_{sit} absorbed into the latent error.

[16]I report AMEs as the percentage-point change in failure probability induced by a 1% change in x (e.g., a change in log compensation); the scaling is for interpretation only.

[17]The static program is

$$\max_{w_{it}} \quad \mathbb{E}_{bt}[A_{it}]w_{it}^{\alpha} - \kappa(1 - \lambda_{\tau_{it}})w_{it} - \chi_b \Pr(\text{SOP fail}_{it}).$$

The resulting first-order condition yields the following first-order approximation of the static Board-optimal log wage level

$$(1-\alpha)\log w_{it} \approx \log\frac{\alpha}{\kappa} + \log \mathbb{E}_{bt}[A_{it}] - \log(1-\lambda_{\tau_t}) - \frac{\chi_b}{\kappa(1-\lambda_{\tau_t})}\frac{\partial \Pr(\text{SOP fail}_{it} \mid \mathcal{I}_{it})}{\partial w_{it}}. \tag{27}$$

The Board's decision is driven by three terms: beliefs about CEO ability, a tenure-specific term, and how the Board internalizes (changes in) Pr(SOP fail) from increasing pay.

The following regression provides a reduced-form description of the Board's decision

$$\log w_{it} = b_1 \mathbb{1}[\text{SOP fail}_{it}] + b_2 \tau_{it} + \mu_i^b + \epsilon_{it}^b, \tag{28}$$

where μ_i^b is a CEO fixed effect and τ_{it} is the CEO's tenure. Because the skill parameters are identified, average log compensation pins down λ_0. The coefficient on tenure, $\hat{b}_2$, helps most in identifying λ_1: conditional on the vote result and the CEO effect, it captures growth in compensation across tenure independent of variation in (beliefs about) ability.

The coefficient on the SOP failure indicator $\hat{b}_1$ is primarily informative about the Board cost χ_b. From (27), χ_b scales the marginal discipline term: holding fixed shareholders' voting rule, a higher χ_b induces the Board to lower compensation more strongly in states with greater failure risk.[18] I also match the variance of the CEO fixed effect, which provides information about variability in both CEO ability and Board beliefs (in particular, σ_{z_b}). Lastly, the log wage autocovariance $\text{Cov}(\log w_{it}, \log w_{it-1})$ helps identify c_w.

Joint identification of χ_s and χ_b. The costs χ_s and χ_b are separated by combining the shareholder voting equation with the Board wage equation. The shareholder probit disciplines how pay and performance map into the probability of SOP failure. In particular, the unconditional failure rate and the sensitivity of failure to compensation are primarily informative about χ_s: a higher shareholder cost of dissent lowers the propensity to fail the vote and dampens the

The expression in the text results from taking first-order conditions, re-expressing the optimal w_{it} in log terms and making the (approximate) substitution $\log(1+x) \approx x$ for small $\frac{\chi_b}{\kappa(1-\lambda_{\tau_t})}\frac{\partial \Pr(\text{SOP fail}_{it})}{\partial w_{it}}$.

[18] Using the realized indicator is appropriate for indirect inference because it is an equilibrium outcome whose conditional expectation equals the ex ante failure probability given $\mathcal{I}_{it}$; since w_{it} is chosen with respect to $\mathcal{I}_{it}$, the wage-failure covariance, and hence the projection coefficient $\hat{b}_1$, summarizes how compensation co-moves with the threat of SOP failure. This moment therefore helps to identify χ_b.

response of failure probabilities to pay.

Holding the shareholder voting rule fixed, the Board wage equation disciplines χ_b. A higher Board cost of failure increases the value of avoiding dissent and therefore strengthens the Board's incentive to compress pay in states with high failure risk. Thus, χ_s is primarily disciplined by the level and slope of the voting rule, while χ_b is primarily disciplined by how compensation co-moves with the equilibrium threat of failure.

Identifying private information. Separately identifying Board and shareholder private information is important for establishing the degree of information asymmetry in the SOP game. On the shareholder side, Appendix IA1.2 shows that with a dispersed shareholder base who receive correlated private signals, the realized proportion of shareholders voting against is informationally equivalent to the realization of a representative shareholder's signal. An implication is that the vote share in excess of the ex ante probability of failure directly reflects (aggregate) private shareholder information.

Letting N_{it} denote the latent index from the probit in (26), the ex ante probability of SOP failure is $\Phi(N_{it})$ and the realized vote share against can be written as

$$V_{it}^{\text{against}} = \Phi(N_{it} + \tilde{z}_{sit}),$$

where $\tilde{z}_{sit}$ captures innovations in (aggregate) shareholder private information that affect vote outcomes. The excess dissent relative to the ex ante probability of failure,

$$\tilde{V}_{it}^{\text{against}} \equiv V_{it}^{\text{against}} - \Phi(N_{it}) = \Phi(N_{it} + \tilde{z}_{sit}) - \Phi(N_{it}),$$

is highly informative about the σ_{z_s}. Intuitively, the model states that, conditional on the information at time t (and, particularly, the output residual, from the econometrician's perspective), only private signal realizations drive variation in the realized proportion voting against relative to the ex-ante probability of failure. The variance of this difference is proportional to the variance of the Shareholder's aggregate private information. In the estimation, I target the cross-sectional variance $\text{Var}(\tilde{V}_{it}^{\text{against}})$, which is increasing in $\sigma_{z_s}^2$.[19]

[19] A second-order expansion around $\tilde{z}_{sit} = 0$ yields $\Phi(N_{it} + \tilde{z}_{sit}) - \Phi(N_{it}) = \phi(N_{it})\tilde{z}_{sit} - \frac{1}{2}N_{it}\phi(N_{it})\tilde{z}_{sit}^2 + \mathcal{O}(\tilde{z}_{sit}^3)$.

On the Board side, the learning structure implies that posterior means separate across CEO tenure. After τ periods, the posterior variance of CEO ability is

$$\sigma_a^2(\tau) = \left(\sigma_0^{-2} + \tau K\right)^{-1}, \qquad K \equiv \sigma_y^{-2} + \sigma_{z_s}^{-2} + \sigma_{z_b}^{-2}.$$

The variance of posterior means is therefore

$$\mathrm{Var}(\mu_{i,\tau}) = \sigma_0^2 - \sigma_a^2(\tau),$$

so types separate over tenure as signals accumulate.

To discipline this learning process in the data, I use the cross-sectional dispersion of a time-varying wage component. Specifically, I first estimate the auxiliary wage equation in (28) and construct the residualized wage $\tilde{w}_{it} = \log w_{it} - \hat{b}_1 \mathbb{1}[\text{SOP fail}_{it}] - \hat{b}_2 \tau_{it}$, which is a reduced-form proxy for the Board's posterior assessment of CEO ability, net of the average tenure profile and the state-contingent SOP component. I then compute, for each tenure value τ, the cross-sectional variance

$$S_\tau^2 = \mathrm{Var}(\tilde{w}_{it} \mid \tau_{it} = \tau),$$

and estimate the tenure-gradient in dispersion:

$$S_\tau^2 = v_0 + v_1 \tau + u_\tau. \tag{29}$$

The moment $\hat{v}_1$ captures the rate at which wage-implied beliefs separate across CEO tenure. Because $\left(\sigma_0^2, \sigma_y^2, \sigma_{z_s}^2\right)$ are disciplined by the output and shareholder-vote moments, this tenure-gradient in wage dispersion is primarily informative about $\sigma_{z_b}^2$. Intuitively, more precise Board signals cause Board beliefs, and therefore Board-selected wages, to separate more quickly across CEO types.

Consequently, $\mathrm{Var}(\tilde{V}_{it}^{\text{against}} \mid N_{it}) = \phi(N_{it})^2 \sigma_{z_s}^2 + \mathcal{O}(\sigma_{z_s}^4)$, and hence $\mathrm{Var}(\tilde{V}_{it}^{\text{against}})$ is proportional to $\sigma_{z_s}^2$ up to the factor $\mathbb{E}[\phi(N_{it})^2]$.

4.1.3. Correlations between Targeted Moments

Table A.1 Panel B reports pairwise correlations across the targeted moments. The moments that discipline the Board and shareholder costs come from distinct conditional projections and are not collinear. Although higher pay predicts a higher probability of SOP failure in the shareholder probit, the unconditional failure component (s_0, column 6) is only moderately correlated with average log compensation (column 9; corr ≈ 0.17). The Board's pay response to failure (b_1, column 10) is essentially orthogonal to the unconditional failure rate (corr ≈ 0.05) and to average pay (corr ≈ -0.03), consistent with b_1 capturing state-contingent pay effects rather than a level effect.

Most importantly for separating χ_s and χ_b, the SOP failure-wage sensitivity (s_1, column 7) and the pay gap across failure states (b_1, column 10) are not highly correlated (corr ≈ 0.40). As argued above, (26) disciplines how failure likelihood varies by wages and performance, and loads primarily on χ_s, while (28) disciplines the Board's state-contingent wage response and loads primarily on χ_b. These moments are not redundant and provide the necessary variation for jointly identifying (χ_s, χ_b).

4.2. Estimated Model Fit

4.2.1. Moment-Matching Exercise

Table 2 displays the closeness of moments from the observed and simulated data.

Table 2 about here

Output and CEO skill moments. Both scale moments (average output and the cost share moment) are well-matched. The model misses the elasticity of output to the wage level (moment 3, y_1: 0.284 in the observed vs. 0.219 in the simulated data). I interpret this moment as a reduced-form elasticity rather than as a direct estimate of the structural curvature α. In the model, wages are chosen dynamically from beliefs that are updated using past output realizations, so the fixed-effect projection of current output on current wages need not perfectly recover the level of α even in simulated data.[20]

[20] In general, the reduced-form elasticity is likely to be attenuated as wages respond to past signals and output shocks, while current output contains a new productivity realization, so the within-spell projection coefficient can differ from α.

The estimation matches the output variance moments closely. The variance of the CEO fixed effect in output (moment 4) disciplines the cross-sectional dispersion in CEO ability, σ_0, while the variance of the output residual (moment 5) disciplines transitory productivity shocks, σ_y. The close fit of both moments suggests that the model captures both persistent CEO-level heterogeneity and within-spell output innovations. These parameters are important for learning, as they govern how quickly public performance realizations move beliefs about CEO ability.

Shareholder moments. The failure rate (moment 6) is matched closely: 6.1% and 5.6% in the observed and simulated data, respectively. The sensitivity of Pr(SOP fail) to the wage is also well-matched (moment 7, s_1). These are key moments for identifying shareholder preferences concerning vote outcomes, so the closeness is reassuring.

The sensitivity of Pr(SOP fail) to output innovations matches the sign in the simulated data, but are large in magnitude relative to the observed data. The model's parsimony may ascribe too much predictive power to these moments. Lastly, the excess vote against variance (moment 9), is closely matched, suggesting that σ_{z_s} is identified.

Board moments. Overall, the Board moments are matched closely. Average log compensation (moment 10) is 1.659 in the data versus 1.715 in simulation, indicating that baseline pay levels, and thus baseline capture λ_0 conditional on the output block, are well disciplined. The within-CEO pay gap across SOP outcomes (moment 11, b_1) is also relatively close (0.334 observed versus 0.266 simulated). This gap is a key statistic for the Board's discipline cost χ_b.

The tenure slope of compensation (moment 12, b_2) is well matched (0.038 observed versus 0.027 simulated), supporting identification of the tenure profile of the governance wedge (i.e., λ_1). The autocovariance of log pay (moment 15) is somewhat higher in simulation (0.681) than in the data (0.594); the model generates slightly too much wage persistence. The cross-sectional variance of the CEO fixed effect in the Board equation (moment 13, 0.571 observed versus 0.600 simulated), and the tenure-gradient dispersion moment (moment 14, 0.008 versus 0.011) are both closely matched, suggesting the model captures heterogeneity in Board beliefs.

4.2.2. Parameter–Moment Sensitivity

I assess local identification using the moment-sensitivity diagnostic in Andrews et al. (2017). Table A.2 shows standardized sensitivities (ranging from −1 to 1) for the parameters describing the Board-shareholder game. The Board's failure cost χ_b is disciplined by b_1, the pay gap across failure states in the Board regression (28).[21] By contrast, the Shareholder's failure cost χ_s is pinned down by SOP fail rate and the probit-response to log pay in (26). The two private-signal noise parameters are also separately identified. The Shareholder signal noise σ_{z_s} is disciplined by s_2, the probit-response to the output shock and $\mathrm{Var}\left(\tilde{V}^{\text{against}}\right)$, as predicted.[22] The Board signal noise σ_{z_b} is instead disciplined by the Board-learning and wage-informativeness moments, most directly the variance of the CEO fixed effect and v_1 (the tenure gradient in the dispersion of wage fixed effects). The level of the governance wedge λ_0 is identified by average log compensation $\mathbb{E}[\log w]$, as predicted, and the tenure profile λ_1 is identified primarily by the tenure coefficient b_2 in the Board regression (28).[23]

4.2.3. Model validation

Confidence in the model's implications will be strengthened if the simulated and observed data produce similar results when examining features of the data not targeted during estimation.

Dynamics of CEO pay around Say-on-Pay failures. Figure 2 displays the dynamics of CEO pay around SOP failure using a difference-in-differences specification, with SOP failure as the "treatment." The model matches the empirical pattern: pay is elevated in the year of failure and then declines in the following year. The Board sets compensation before the vote, using its private signal about CEO ability. A failure year is therefore a year in which the Board chooses a relatively high wage, but shareholders subsequently view the wage as too high given their own information and realized performance. After the failed vote, dissent

[21] Note that χ_b can also load on the tenure-gradient in wage dispersion, v_1, because the moment is computed from equilibrium wages rather than directly observed beliefs. A larger χ_b changes the mapping from Board beliefs to wages by inducing pay compression in high-failure-risk states. Thus, v_1 is not a pure signal-precision moment, although it is informative about the precision of the Board's private signal.

[22] Note that σ_{z_s} is also sensitive to moments from the Board regression. Shareholder signal precision ($1/\sigma_{z_s}$) governs how informative a failed vote is for the Board's posterior, so the Board's anticipated pay response to failure (b_1) depends on σ_{z_s}.

[23] λ_1 also responds to $\mathrm{Var}\left(\mu^b\right)$ and v_1, both of which reflect tenure-dependent variation in Board beliefs that the tenure profile of the wedge helps absorb.

reveals negative information about CEO type, leading the Board to revise beliefs downward and keep pay reduced in subsequent years.

Figure 2 about here

Changes in CEO pay Following Say-on-Pay disapproval. Figure 3 displays how the observed and simulated data compare in terms of changes in CEO compensation (from t to $t + 1$) in response to different SOP disapproval rates (in t). The slope of this relation is very similar in the model and data (though understandably more precise in the model). A key prediction of the model is that SOP disapproval conveys that shareholders believe the CEO is low-type. The figure shows that the Board endogenously internalizes shareholder beliefs into future pay decisions, and that the relation between the revision in pay and the magnitude of the adjustment downwards (as proxied by vote realization) is closely matched in the observed and simulated data.

Figure 3 about here

5. Results

This section presents the results from the model estimation. I first discuss the estimated structural parameters and their economic magnitudes. I then discuss the impact of Say-on-Pay on compensation and shareholder value.

5.1. Estimated Structural Parameters and Economic Implications

Table 3 displays the estimated parameters. To aid interpretation, all unbounded parameters are presented as a percentage of shareholder value.[24]

Table 3 about here

[24] In Appendix IA1.7, I derive the model's benchmark measure of average shareholder value used for normalization. All qualitative and directional conclusions are unchanged under alternative normalizations (e.g., by average operating profits or average CEO compensation); for brevity, I report only the shareholder-value normalization in Table 3.

5.1.1. The Board and Shareholder Costs to Say-on-Pay Failure

I estimate that the Board behaves as if SOP failure imposes a utility cost of 1.622% of average shareholder value. For shareholders, this cost is 1.395% of value (i.e., of a shareholder's equity stake). It is important to emphasize that these are utility costs: SOP failure does not affect value directly, but the Board and shareholders must behave as if it does for the model to match observed outcomes. In particular, a positive χ_s rationalizes why shareholders pass some SOP votes that would fail absent this cost.

These estimates highlight a central mechanism. Nonbinding SOP functions as a disciplinary technology, a form of shareholder voice, in which shareholders can punish perceived overpayment through dissent, but internalize a cost from doing so. This interpretation is consistent with evidence that negative voting outcomes, or the threat of them, can influence corporate policies even when votes are nonbinding (e.g., Del Guercio et al., 2008), and with the broader view that shareholder intervention is costly (e.g., Gantchev, 2013). Further, Edmans et al. (2023) provide survey evidence of this cost: institutional investors express reluctance to fail SOP votes because dissenting from management on a prominent policy is costly.[25] Importantly, the model implies that SOP's effects operate largely through equilibrium anticipation: the compensation policy brought to a vote is itself endogenous, chosen to manage the ex ante threat of failure, helping reconcile why SOP votes typically pass with high support.

5.1.2. Economic Implications of Other Parameters

Governance wedge. The estimates imply a positive Board–shareholder wedge in pay-setting. The underlying parameters are $\lambda_0 = 0.303$ and $\lambda_1 = 0.030$, so the wedge λ_τ rises with CEO tenure. To interpret the magnitude, I translate λ_τ into the implicit weight the Board places on CEO pay relative to shareholder profits:

$$\nu_\tau = \frac{\lambda_\tau}{1 + \lambda_\tau}.$$

[25]See Online Appendix A. Investors also feel that dissent may create future monitoring costs, as Boards will engage with shareholders repeatedly about changing compensation policy.

This weight is approximately 0.23 at tenure zero, 0.29 at ten years of tenure, and 0.35 by year twenty; the model estimates a modest but increasing governance wedge over the CEO's tenure. These magnitudes are in the range of estimates from the executive compensation literature. Taylor (2013) finds that CEOs capture roughly half of the surplus from good news, while Barry et al. (2025) estimate pure managerial bargaining power of about 0.26.[26]

Output and CEO skill parameters. The volatility of CEO skill is 1.185% of shareholder value, implying that CEO skill matters (supporting Taylor (2010), who finds that CEO fixed effects matter greatly). The output shock volatility σ_y is 1.405% of shareholder value: a large portion of observed variation in output comes from randomness, rather than CEO skill. The curvature of CEO-driven revenues with respect to compensation is $\alpha = 0.463$. This reduced-form curvature is consistent with span-of-control views in which managerial services scale output concavely, or incentive models in which higher pay induces greater managerial input but with diminishing marginal output gains.[27]

Private information. The standard deviations of the Board's and shareholders' private signals are 1.314% and 1.386% of shareholder value, respectively. The Board's signal innovation is slightly less volatile than the shareholders' signal innovation. In addition, the Board observes its signal prior to choosing compensation; therefore, at the strategy stage the Board's posterior beliefs are more precise. Figure A.1 plots the evolution of posterior belief volatility over tenure and shows that learning is somewhat gradual: substantial uncertainty remains through the median CEO tenure length in the sample ($\tau = 7$, Table 1).

5.2. How Much Does Say-on-Pay Impact Compensation and Value?

The estimated Board cost to SOP failure implies that SOP can discipline compensation policy, but its magnitude is not directly interpretable in terms of pay and value. To quantify the

[26]In Barry et al. (2025), a manager's share of surplus rises over tenure due to labor market competition. The present model abstracts from explicit competition and outside offers, so λ_τ should be interpreted as a reduced-form governance wedge rather than pure agency. In particular, some degree of "capture" could be privately optimal for shareholders as part of an efficient retention or incentive arrangement for high-skill CEOs; these considerations are outside the model in this paper.

[27]For span-of-control foundations in which managerial talent/services scale firm output concavely, see Lucas (1978) and Rosen (1982). For executive-compensation models linking pay to managerial actions and firm value (assignment and moral-hazard incentives), see the survey Edmans and Gabaix (2016).

impact of SOP, I simulate a counterfactual economy in which the Board places no weight on SOP failure by setting $\chi_b = 0$ and holding all other parameters at estimated values.

Table 4 about here

Table 4 reports the resulting changes in compensation and value. On average, removing SOP increases total pay by 4.77%, implying that SOP reduces compensation by this amount. This magnitude is broadly in line with Correa and Lel (2016), who find that total CEO pay falls by roughly 7% upon SOP adoption in a cross-country setting. There is meaningful heterogeneity in the pay response. SOP has a larger effect for below-average ability CEOs (7.42% versus 3.84%), for CEOs early in tenure (5.25% versus 2.85% later in tenure), and in observations that fail SOP in the baseline economy (8.42% versus 1.11% for pass), consistent with discipline being strongest when the threat of dissent is most salient.

The model implies that SOP increases shareholder value by 2.22% on average, with heterogeneity that mirrors where discipline is most consequential: value gains are largest in baseline-fail observations (4.22%) and for below-average ability CEOs (3.40%), with later-tenure CEOs also benefiting more than early-tenure CEOs (2.46% versus 2.15%). Cuñat et al. (2016) estimate that adopting SOP increases market value by roughly 5% using close adoption votes prior to Dodd–Frank; since their estimate is local to firms at the adoption margin, it is reassuring that the model's average value effect is smaller.

The contribution of this counterfactual relative to the empirical adoption evidence (Cuñat et al., 2016; Correa and Lel, 2016) is the mechanism. In the model, SOP's effect is not generated by a fixed compensation policy evaluated by shareholders; rather, the compensation policy brought to a vote is itself endogenous. Anticipating shareholders' voting rule and the possibility of failure, the Board adjusts compensation to manage ex ante failure risk. In other words, the proposal put forth is endogenous to the equilibrium stipulated by the voting environment. This implies that high observed pass rates are consistent with discipline: SOP operates primarily through the threat of failure and the induced selection of policies likely to pass, not through frequent realized failures. Observed failure realizations are therefore an incomplete statistic for SOP's impact; the relevant object is the ex ante failure risk that the Board internalizes when setting compensation.

6. Counterfactual Analysis

Section 5.2 establishes that SOP already disciplines pay through the Board's anticipation of failure: the intervention channel of the vote. The communication channel, or the within-period transmission of shareholder information into pay-setting, is left indirect by the current advisory format. A failed vote does not change the implemented wage, so vote-revealed information enters pay only across periods. This section asks how much additional shareholder value is available if the regime instead routes vote-revealed information directly into within-period pay-setting.

6.1. Design and mechanism

This subsection gives a conceptual outline of the counterfactual; the technical description is in Appendix B. As in Section 3.4, the Board and Shareholder come into period t with the shared prior $a \sim N(\mu_{at}, \sigma_{at}^2)$. The vote moves from the end of the period to within-period: the Board observes its private signal z_{bt} and proposes a wage w_t^{pass}, the Shareholder receives its signal z_{st} and votes, and the implemented wage is determined by the vote outcome before output is realized. Formally, the implemented wage is

$$w_t = \begin{cases} w_t^{\text{pass}} = \tilde{\omega}(\mu_{at}, \sigma_{at}^2, w_{t-1}) & \text{if SOP pass} \\ w_t^{\text{info}} & \text{if SOP fail} \end{cases}, \tag{30}$$

where w_t^{info} is the wage the Board would choose after observing both its own signal z_{bt} and the shareholder signal z_{st}, while retaining the objective from the baseline (this improves the information available to the Board but does not remove the governance wedge λ_τ).

Figure 4 about here

Figure 4 shows how the counterfactual changes the structure of the vote. Because failure now affects within-period implementation, shareholders choose vote sensitivity s_t based on the pass–fail payoff difference rather than on a fixed expected utility cost, and the Board selects w_t^{pass} anticipating this response. The vote thus becomes a within-period communication device: dissent directly translates shareholder information into the implemented wage.

6.2. Results

Table 5 about here

Table 5 shows the results from the counterfactual. I compare wage levels and shareholder value to the advisory baseline. Strengthening the within-period flow of shareholder information into pay-setting lowers average CEO pay by 3.73% and raises shareholder value by 1.28% (Table 5). The SOP failure rate rises from 5.50% in the baseline to 9.79% under the counterfactual, reflecting that within-period failure is more attractive to shareholders when it directly impacts pay.

The cross-sectional splits in Table 5 show where the gains concentrate. Value gains are largest for low-ability CEOs (+1.70%) and for observations with high predicted baseline failure (+1.83%), consistent with the counterfactual being most valuable where the baseline vote is most informative about overpayment. Pay declines are largest for low-ability CEOs (−4.58%) and for observations with low predicted baseline failure (−5.88%), while pay falls by 3.19% for high predicted-failure observations. Thus, the largest value gains need not occur in the cells with the largest pay declines: the value effect depends on both the wage reduction and how far the baseline wage is from the information-pooled benchmark.

These gains reflect the strength of the information channel built into the counterfactual. To quantify how shareholder welfare varies with this channel, I introduce an information-pass-through friction $\rho \in [0, 1]$, which governs how much of the shareholder information revealed by a failed vote is incorporated into the revised wage. At $\rho = 0$, failure fully implements the information-pooled wage w_t^{info}. At $\rho = 1$, the failure-state wage equals the pass-state wage, so failure has no within-period effect on pay.

Solving the model at $\rho \in \{0, 0.25, 0.5, 0.75, 1\}$ shows that the gains are increasing in information pass-through, $1 - \rho$. The full-information case, $\rho = 0$, matches the baseline counterfactual effects. while the pay effect shrinks to zero as ρ approaches one, the advisory baseline.[28]

[28] The $\rho = 1$ case corresponds to the advisory implementation rule within the counterfactual timing: failure does not change the within-period wage. It is therefore the natural no-pass-through benchmark for this exercise, but it is not mechanically identical to the baseline advisory model because the counterfactual re-solves the equilibrium under the revised timing and information structure. The small positive value effect reported at $\rho = 1$ is economically negligible and reflects numerical approximation from re-solving and simulating the counterfactual equilibrium.

Thus, the full-information counterfactual provides an upper bound on the shareholder return available from strengthening the communication channel, while intermediate values quantify the value created by partial pass-through of vote-revealed information into pay-setting.

The counterfactual can be interpreted as a revise-and-resubmit version of SOP. A failed vote not only communicates dissatisfaction; it triggers a revised compensation decision that incorporates shareholder information. This makes the vote a within-period information-transmission device, related to models of shareholder communication with management (e.g., Kakhbod et al., 2023; Levit, 2020). In contrast to settings where intervention can undermine communication, the counterfactual maps the information in a failed vote directly into pay: conditional on failure, compensation is revised to w_t^{info}.

As a regime change, this counterfactual is subject to the Lucas critique. I re-solve shareholders' voting strategy under the counterfactual rule, holding fixed the governance wedge λ_τ, the failure-cost parameters (χ_b, χ_s), and the underlying shareholder-side primitives governing the aggregated voting rule. A rule that makes failure immediately consequential would likely change these objects: failure would trigger compensation revisions, additional engagement, and more governance work for both boards and institutional shareholders. These results should be interpreted as arising from adding within-period responsiveness under baseline preferences, not as a full equilibrium prediction of how such a regime would operate after preferences and stewardship policies adjust.

7. Conclusion

Say-on-Pay (SOP) is a prominent shareholder voice mechanism directed at executive compensation, yet its nonbinding nature and high support rates leave its impact as a governance mechanism ambiguous. This paper clarifies both the magnitude and mechanism of SOP's impact. I estimate a structural model in which the Board and shareholders can differ in preferred pay levels and in their aversion to failed SOP votes, and use the estimated model to evaluate a counterfactual vote design that strengthens SOP's within-period communication channel.

In equilibrium, the Board sets pay anticipating how shareholders translate realized compensation and information into dissent. Although only about 6% of votes fail, the estimated model implies that the disciplinary channel of SOP lowers CEO pay by 4.77% and increases shareholder value by 2.22% on average. The key implication is that the proposal put forth is endogenous to the voting environment. High observed pass rates are consistent with discipline because shareholder voting can operate primarily through the threat of failure and the induced selection of proposals likely to pass, rather than through frequent realized failures. This logic applies broadly to other corporate policies subject to shareholder approval, such as equity issuance or major transactions.

I then use the estimated model to evaluate a counterfactual vote design that lets a failed vote feed shareholder information directly into within-period pay-setting. Relative to the baseline advisory regime, pay would fall by an additional 3.73% and shareholder value rises by an additional 1.28%. The mechanism is that tying the failure-state wage to a posterior that pools Board and shareholder signals makes the vote a within-period communication device, complementing the discipline already operating in the baseline regime.

Future research could integrate a strategic voting game among heterogeneous shareholders with policy-setting by the corporate decision-maker, making both the probability of failure and the informational content of the vote equilibrium objects. This would help quantify how, for example, coordination among shareholders or pivotality effects amplify or mute the vote's disciplinary effect.

References

Aggarwal, R., Dahiya, S., and Prabhala, N. R. (2019). The power of shareholder votes: Evidence from uncontested director elections. *Journal of Financial Economics*, 133(1):134–153.

Andrews, I., Gentzkow, M., and Shapiro, J. M. (2017). Measuring the sensitivity of parameter estimates to estimation moments. *The Quarterly Journal of Economics*, 132(4):1553–1592.

Armstrong, C. S., Gow, I. D., and Larcker, D. F. (2013). The efficacy of shareholder voting: Evidence from equity compensation plans. *Journal of Accounting Research*, 51(5):909–950.

Barry, J. W., Lyman, N., and Zhao, L. (2025). Human capital, competition and mobility in the managerial labor market.

Barry, J. W., Pinnington, J., and Zhao, L. (2026). Information and preferences in shareholder voting.

Bebchuk, L. A. and Fried, J. M. (2003). Executive compensation as an agency problem. *Journal of economic perspectives*, 17(3):71–92.

Becht, M., Polo, A., and Rossi, S. (2016). Does mandatory shareholder voting prevent bad acquisitions? *The Review of financial studies*, 29(11):3035–3067.

Bertrand, M. and Mullainathan, S. (2001). Are CEOs rewarded for luck? The ones without principals are. *The Quarterly Journal of Economics*, 116(3):901–932.

Blonien, P., Crane, A. D., Crotty, K., and De Angelis, D. (2025). Proxy advice and errors in shareholder voting.

Cai, J. and Walkling, R. A. (2011). Shareholders' say on pay: Does it create value? *Journal of Financial and Quantitative Analysis*, 46(2):299–339.

Coles, J. L., Daniel, N. D., and Naveen, L. (2014). Co-opted boards. *The Review of Financial Studies*, 27(6):1751–1796.

Correa, R. and Lel, U. (2016). Say on pay laws, executive compensation, pay slice, and firm valuation around the world. *Journal of Financial Economics*, 122(3):500–520.

Cuñat, V., Gine, M., and Guadalupe, M. (2012). The vote is cast: The effect of corporate governance on shareholder value. *The Journal of Finance*, 67(5):1943–1977.

Cuñat, V., Giné, M., and Guadalupe, M. (2016). Say pays! Shareholder voice and firm performance. *Review of Finance*, 20(5):1799–1834.

Del Guercio, D., Seery, L., and Woidtke, T. (2008). Do boards pay attention when institutional investor activists "just vote no"? *Journal of Financial Economics*, 90(1):84–103.

Dey, A., Starkweather, A., and White, J. T. (2024). Proxy advisory firms and corporate shareholder engagement. *The Review of Financial Studies*, 37(12):3877–3931.

Edmans, A. and Gabaix, X. (2016). Executive compensation: A modern primer. *Journal of Economic Literature*, 54(4):1232–1287.

Edmans, A., Gosling, T., and Jenter, D. (2023). CEO compensation: Evidence from the field. *Journal of Financial Economics*, 150(3):103718.

Ferri, F. and Maber, D. A. (2013). Say on pay votes and CEO compensation: Evidence from the UK. *Review of Finance*, 17(2):527–563.

Fisch, J., Palia, D., and Solomon, S. D. (2018). Is say on pay all about pay: The impact of firm performance. *Harvard Business Law Review*, 8:101.

Flammer, C. (2015). Does corporate social responsibility lead to superior financial performance? A regression discontinuity approach. *Management science*, 61(11):2549–2568.

Fos, V. (2017). The disciplinary effects of proxy contests. *Management Science*, 63(3):655–671.

Fos, V. and Tsoutsoura, M. (2014). Shareholder democracy in play: Career consequences of proxy contests. *Journal of Financial Economics*, 114(2):316–340.

Gantchev, N. (2013). The costs of shareholder activism: Evidence from a sequential decision model. *Journal of Financial Economics*, 107(3):610–631.

Gantchev, N., Gredil, O. R., and Jotikasthira, C. (2019). Governance under the gun: Spillover effects of hedge fund activism. *Review of Finance*, 23(6):1031–1068.

Gillan, S. L. and Starks, L. T. (2007). The evolution of shareholder activism in the United States. *Journal of Applied Corporate Finance*, 19(1).

Heath, D., Macciocchi, D., and Ringgenberg, M. C. (2026). The economics of investor engagement.

Hermalin, B. E. and Weisbach, M. S. (1998). Endogenously chosen boards of directors and their monitoring of the CEO. *American Economic Review*, pages 96–118.

Hirschman, A. O. (1970). *Exit, voice, and loyalty: Responses to decline in firms, organizations, and states*, volume 25. Harvard University Press.

Holderness, C. G. (2018). Equity issuances and agency costs: The telling story of shareholder approval around the world. *Journal of Financial Economics*, 129(3):415–439.

Iliev, P., Lins, K. V., Miller, D. P., and Roth, L. (2015). Shareholder voting and corporate governance around the world. *The Review of Financial Studies*, 28(8):2167–2202.

ISS (2022). United States proxy voting guidelines. https://www.issgovernance.com/file/policy/active/americas/US-Voting-Guidelines.pdf [Accessed: 2023-02-25].

Johnson, T. L. and Swem, N. (2021). Reputation and investor activism: A structural approach. *Journal of Financial Economics*, 139(1):29–56.

Kakhbod, A., Loginova, U., Malenko, A., and Malenko, N. (2023). Advising the management: A theory of shareholder engagement. *The Review of Financial Studies*, 36(4):1319–1363.

Kaplan, S. N. (2013). CEO pay and corporate governance in the US: Perceptions, facts, and challenges. *Journal of Applied Corporate Finance*, 25(2):8–25.

Levit, D. (2019). Soft shareholder activism. *The Review of Financial Studies*, 32(7):2775–2808.

Levit, D. (2020). Words speak louder without actions. *The Journal of Finance*, 75(1):91–131.

Levit, D. and Malenko, N. (2011). Nonbinding voting for shareholder proposals. *The Journal of Finance*, 66(5):1579–1614.

Lucas, Robert E., J. (1978). On the size distribution of business firms. *The Bell Journal of Economics*, 9(2):508–523.

McFadden, D. (1989). A method of simulated moments for estimation of discrete response models without numerical integration. *Econometrica*, pages 995–1026.

Page, T. B. (2018). CEO attributes, compensation, and firm value: Evidence from a structural estimation. *Journal of Financial Economics*, 128(2):378–401.

Rosen, S. (1982). Authority, control, and the distribution of earnings. *The Bell Journal of Economics*, 13(2):311–323.

Shleifer, A. and Vishny, R. W. (1997). A survey of corporate governance. *The Journal of Finance*, 52(2):737–783.

Taylor, L. A. (2010). Why are CEOs rarely fired? Evidence from structural estimation. *The Journal of Finance*, 65(6):2051–2087.

Taylor, L. A. (2013). CEO wage dynamics: Estimates from a learning model. *Journal of Financial Economics*, 108(1):79–98.

Wagner, A. F. and Wenk, C. (2025). Binding say-on-pay and shareholder value. *The Review of Corporate Finance Studies*, page cfaf010.

Zhu, C. (2021). The preventive effect of hedge fund activism: Investment, CEO compensation and payout policies. *International Journal of Managerial Finance*, 17(3):401–415.

Figure 1. Model timeline

This figure displays the within-period model timeline. The top timeline displays the timeline as it maps to practice; the bottom as it maps to the sequencing of events within each period t. Figure IA.1 displays an in-depth timeline which incorporates the timing of the strategies by the Board and Shareholder (See Appendix IA1.5).

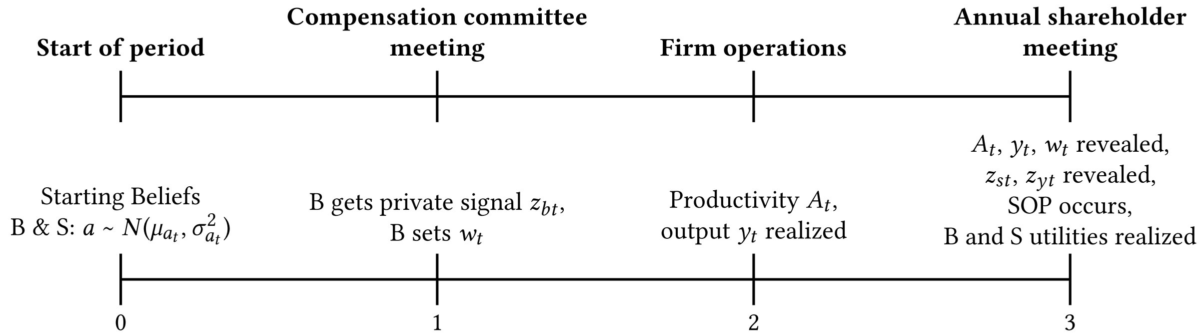


Figure 2. Model fit: Compensation dynamics around SOP failure

This figure displays compensation dynamics around the first SOP failure in both the observed and simulated data. In each dataset, I estimate the following event-study specification:

$$\log w_{it} = \iota_i + \sum_{k=-4,\, k\neq-2}^{4} \beta_k \mathbf{1}\left\{\left(t - T_i^{fail}\right) = k\right\} + \varepsilon_{it},$$

where i indicates a CEO spell, t indicates event time, ι_i is a CEO-spell fixed effect, and T_i^{fail} is the first SOP failure in the CEO spell. The omitted category is $k = -2$, two years before the SOP failure. In the observed data regression, I include calendar-year fixed effects to account for aggregate shocks and trends; in the simulated data, I omit calendar-year fixed effects. Each point reports the estimated difference in log compensation at event time k relative to $k = -2$.

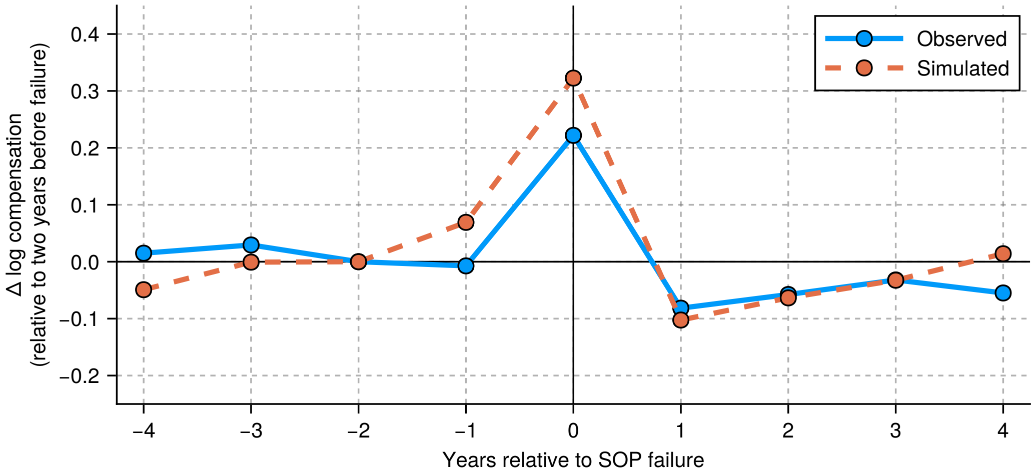


Figure 3. Model fit: The relation between changes in CEO pay and SOP vote outcomes
This figure displays the relation between changes in CEO pay and the SOP vote outcome in both the observed and simulated data. In both panels, I display a binned scatterplot estimated from a regression of the (residualized) change in CEO pay on SOP disapproval, expressed as (residualized) distance to the predicted vote probability, given the state at time t. The data regression includes a year fixed; both regressions include CEO-spell fixed effects and a control for the output shock from the previous period.

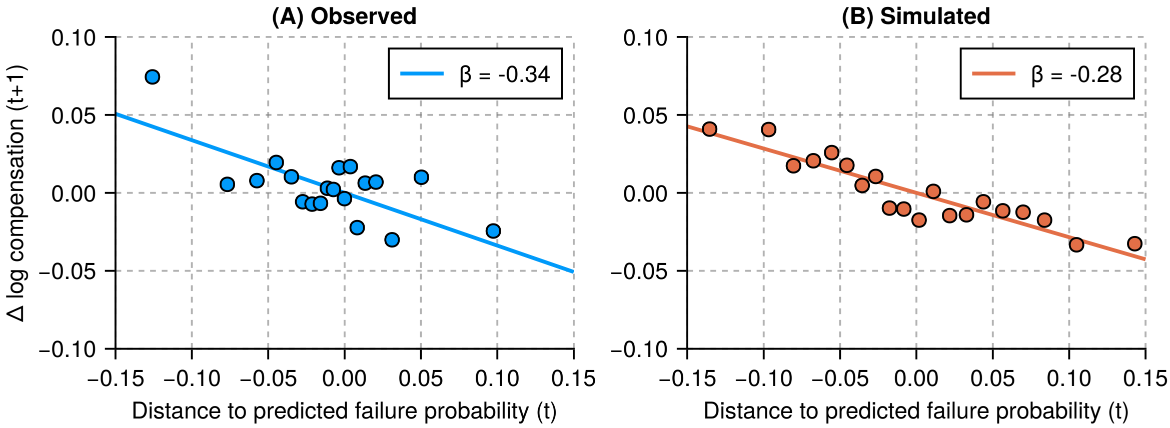


Figure 4. Counterfactual model timeline

This figure displays the within-period model timeline of the full-information counterfactual. The vote moves from the end of the period (baseline) to within-period, and a failed vote implements w_t^{info}, the wage that is optimal under the posterior pooling Board and Shareholder signals.

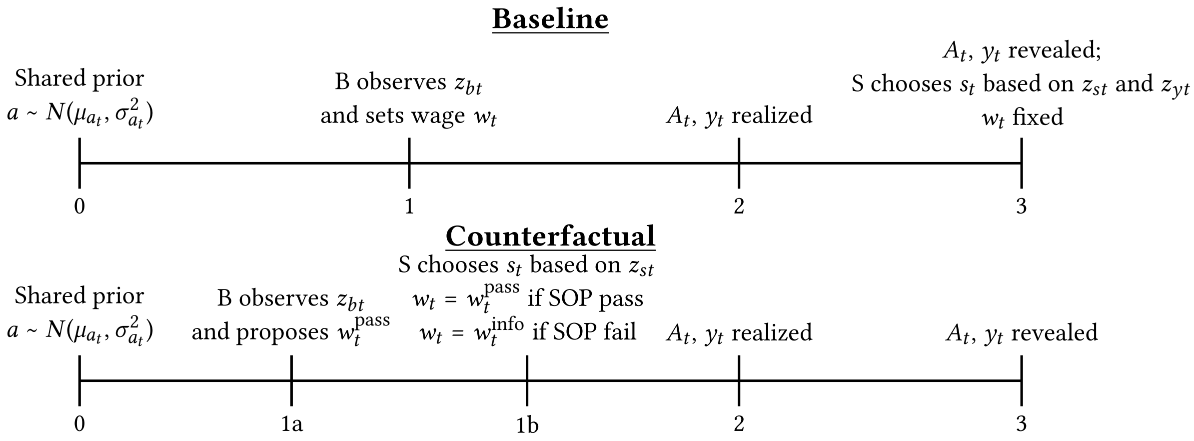


Table 1. Summary statistics

This table reports descriptive statistics for the variables used in the model estimation. The sample merges Compustat, Execucomp, and ISS for the fiscal years 2011–2024. All dollar variables are deflated to 2018 dollars. "Vote against SOP" is the fraction of votes cast against the Say-on-Pay proposal. "SOP failure" and the other indented rows are indicators equal to one when the vote-against share exceeds 30%, 20%, and 50%, respectively. "Total compensation" is total expected pay (in millions, i.e. TDC1). "Current compensation share" is current compensation divided by total compensation. "Change in total pay" is the change in (log) total compensation from the prior fiscal year, within CEO spell. "CEO tenure" is years since the CEO assumed office, and "Length of CEO tenure" is the total duration of the CEO's spell at the firm (one observation per CEO–firm spell). "Assets", "Revenues", and "Market capitalization" are in billions; market capitalization is measured at the end of the fiscal year. "Return on assets" is operating income divided by assets, and "Profit margin" is operating income divided by sales.

Variable	Unit	N	Mean	Std dev	25%	50%	75%
Say-on-Pay							
Vote against SOP (%)	%	13,109	0.082	0.099	0.025	0.046	0.085
SOP failure: $\mathbb{1}$[More than 30% against]	{0,1}	13,109	0.061				
$\mathbb{1}$[More than 20% against]	{0,1}	13,109	0.105				
$\mathbb{1}$[More than 50% against]	{0,1}	13,109	0.018				
CEO							
Total compensation (TDC1)	$mil	13,109	7.469	7.815	3.041	5.643	9.800
Current compensation share	%	13,109	0.228	0.188	0.111	0.166	0.269
Change in total pay	%	11,659	0.019	0.420	-0.101	0.016	0.177
CEO tenure	years	13,109	6.278	5.733	2	5	9
Length of CEO tenure	years	2,324	8.725	6.159	4	7	12
Firm							
Assets	$bil	13,109	31.392	162.531	1.281	4.084	13.643
Revenues	$bil	13,109	10.819	33.231	0.824	2.323	7.628
Market capitalization	$bil	13,109	18.896	74.511	1.251	3.454	12.044
Return on assets	%	13,109	0.124	0.090	0.068	0.117	0.170
Profit margin	%	13,109	0.208	0.161	0.103	0.175	0.289

Table 2. Model fit

This table displays the estimated model's fit. Following the identification strategy in Section 4.1, I estimate the model in Section 3 by simulated method of moments, as described in Appendix IA2. The table reports the targeted moments in the data and in the simulated model evaluated at the estimated parameter vector. The final row reports the SMM objective value, or J-statistic.

	Description	Notation	Observed	Simulated	Difference
Output & CEO skill					
(1)	Average log output	$\mathbb{E}[\log y]$	7.859	8.016	−0.157
			(0.038)		
(2)	Cost share	k_1	0.366	0.379	−0.012
			(0.586)		
(3)	Elasticity of output to wage	y_1	0.284	0.219	0.065
			(0.017)		
(4)	Output FE variance	$\text{Var}(\mu^y)$	0.450	0.530	−0.080
			(0.019)		
(5)	Output residual variance	$\text{Var}(\epsilon^y)$	0.471	0.481	−0.010
			(0.010)		
Shareholder					
(6)	SOP failure rate	$\mathbb{E}[\text{SOP fail}]$	0.061	0.056	0.005
			(0.003)		
(7)	SOP fail wage sensitivity	s_1	0.058	0.045	0.013
			(0.006)		
(8)	SOP fail output shock sensitivity	s_2	−0.045	−0.077	0.032
			(0.006)		
(9)	Excess vote-against variance	$\text{Var}\left(\tilde{V}^{\text{against}}\right)$	0.119	0.106	0.013
			(0.006)		
Board					
(10)	Average log wage	$\mathbb{E}[\log w]$	1.659	1.715	−0.057
			(0.019)		
(11)	Change in log wage (SOP fail)	b_1	0.334	0.266	0.068
			(0.029)		
(12)	Wage growth over tenure	b_2	0.038	0.027	0.011
			(0.002)		
(13)	Wage FE variance	$\text{Var}\left(\mu^b\right)$	0.571	0.600	−0.030
			(0.016)		
(14)	Wage FE variance tenure slope	v_1	0.008	0.011	−0.003
			(0.002)		
(15)	Log wage autocovariance	$\text{Cov}(\log w, \log w_{-})$	0.594	0.681	−0.087
			(0.021)		
			SMM objective (J-stat) = 97.907		

Table 3. Estimated parameters

This table displays estimates of the parameters that drive the model in Section 3, given the estimation strategy in Section 4. Unscaled parameters are displayed as a percentage of average shareholder value (i.e., % s.v.), with standard errors derived via the delta method, see Appendix IA1.7 for details. The value of each parameter is displayed, along with its standard errors in parentheses. The moment Jacobian underlying the standard errors is computed by central differences averaged across 100 simulation seeds.

Description	Unit	Notation	Value	
Output & CEO skill				
Prior std dev of CEO ability	% s.v.	σ_0	1.185%	(0.120%)
Std dev of productivity shock	% s.v.	σ_y	1.405%	(0.092%)
Output—CEO wage elasticity	(0,1)	α	0.463	(0.036)
Scaling factor (output)	log $mil	$\log \eta$	7.219	(0.074)
Scaling factor (cost)	(0,1)	κ	0.379	(0.025)
Shareholder				
Std dev of Shareholder signal	% s.v.	σ_{z_s}	1.386%	(0.170%)
Shareholder SOP failure cost	% s.v.	χ_s	1.395%	(0.303%)
Board				
Std dev of Board signal	% s.v.	σ_{z_b}	1.314%	(0.074%)
Board SOP failure cost	% s.v.	χ_b	1.622%	(0.171%)
CEO board capture (constant)	(0,1)	λ_0	0.303	(0.061)
CEO board capture (growth)	(0,1)	λ_1	0.030	(0.002)
CEO wage adjustment cost	% s.v.	c_w	2.008%	(0.304%)

Table 4. The impact of Say-on-Pay on compensation and shareholder value

This table analyzes the impact of SOP on compensation policy and shareholder value. I simulate a counterfactual, using the same sequence of shocks as in the baseline estimation, in which the Board cost of SOP failure (χ_b) is set to zero while holding all other parameters fixed. This removes the Board's ex ante incentive to avoid SOP failure, and therefore isolates the disciplinary channel of the vote. The first row reports the average counterfactual percentage change in the wage level; the second reports the change in shareholder value from (22). The first column reports average percentage changes for the full simulated sample. The remaining columns split the sample by cross-sectional characteristics. In columns two and three, "Low Ability" means the CEO's type a is below zero, and "High Ability" means a is above zero. In columns four and five, "Early" refers to the first five years of CEO tenure, while "Late" refers to tenure beyond five years. In columns six and seven, "Low PredFail" and "High PredFail" split observations by the predicted baseline probability of SOP failure, with "High PredFail" corresponding to observations in the top quartile and "Low PredFail" corresponding to the bottom quartile.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
		Ability		**Tenure**		**PredFail**	
	Full sample	Low	High	Early	Late	Low	High
Wage level	+4.77%	+7.42%	+3.84%	+5.25%	+2.85%	+1.11%	+8.42%
Firm value	-2.22%	-3.40%	-1.34%	-2.15%	-2.46%	+0.02%	-4.22%
Pr(Fail), baseline	5.50%	9.29%	2.79%	5.26%	6.27%	0.14%	14.93%

Table 5. Full-information counterfactual: compensation and shareholder value

This table reports counterfactual outcomes from the information-pass-through regime described in Section 6 and Appendix B. In the counterfactual, the Board proposes compensation and shareholders vote within-period, before output is realized. If the vote passes, the CEO receives the proposed wage. If the vote fails, the implemented wage is adjusted toward w_t^{info}, the wage the Board would choose after observing both its own signal and the shareholder signal. I re-solve the model holding all primitives at their estimated values and apply the same sequence of shocks as in the baseline. Panel A reports the full-pass-through case, in which failure fully implements w_t^{info}. The first row reports the average percentage change in the wage level; the second reports the change in shareholder value from (22); and the third and fourth rows report the SOP failure rate in the baseline and counterfactual regimes. The first column shows the full sample. The remaining columns split the sample by cross-sectional characteristics: "Low/High Ability" splits on whether the CEO's type a is below or above zero; "Early/Late" tenure splits at five years; and "Low/High PredFail" splits on the predicted baseline failure probability. Panel B varies the communication-friction parameter ρ: $\rho = 0$ is full pass-through to w_t^{info}, while $\rho = 1$ is the pure advisory implementation rule under which failure has no within-period effect on pay. Because a within-period vote is only relevant once there is a previously approved wage, I drop $\tau = 0$ observations.

Panel A. Cross-sectional effects

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
		Ability		**Tenure**		**PredFail**	
	Full sample	Low	High	Early	Late	Low	High
Wage level	-3.73%	-4.58%	-4.20%	-3.78%	-3.34%	-5.88%	-3.19%
Firm value	+1.28%	+1.70%	+1.27%	+1.19%	+1.53%	+1.83%	+1.04%
Pr(Fail), baseline	5.50%	9.29%	2.79%	5.26%	6.27%	0.14%	14.93%
Pr(Fail), CF	9.79%	14.87%	9.57%	7.03%	18.39%	11.59%	11.71%

Panel B. Impact of communication frictions

ρ (information pass-through friction)	0.00	0.25	0.50	0.75	1.00
Wage level	-3.73%	-2.77%	-1.75%	-0.83%	+0.00%
Firm value	+1.28%	+1.12%	+0.87%	+0.51%	+0.08%

A. Appendix Figures and Tables

Figure A.1. Board and Shareholder volatility of beliefs over CEO tenure
This figure displays the volatility of Board and shareholder beliefs about CEO ability as a function of CEO tenure τ at the time at which each agent plays their strategy, using the parameter estimates from Table 3. Each volatility of beliefs is at the time that each party plays their strategy: the Board's belief volatility is $\sigma_{bt|z_b}$, as in (19), and the Shareholder's belief volatility is σ_{at} (i.e., the prior).

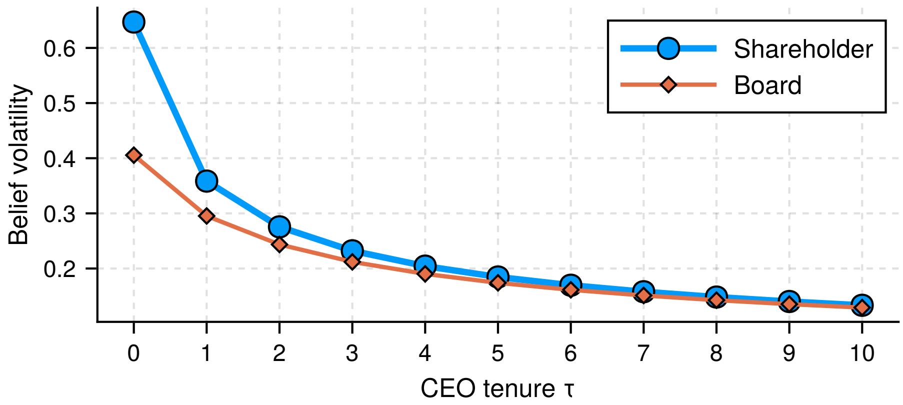


Table A.1. Moment targeting exercise

This table provides detail on targeted empirical moments. Panel A describes each moment, and lists the parameter(s) it targets. Panel B provides pairwise correlations. The correlations in Panel B are constructed via the empirical variance-covariance matrix, estimated using the influence function approach detailed in Appendix IA2.3.

Panel A. Targeted empirical moments

	Description	**Notation**	**Target**
	Output & CEO skill		
(1)	Average log output	$\mathbb{E}[\log y]$	$\log \eta$
(2)	Cost share	k_1	κ
(3)	Elasticity of output to wage	y_1	α
(4)	Output FE variance	$\mathrm{Var}(\mu^y)$	σ_0
(5)	Output residual variance	$\mathrm{Var}(\epsilon^y)$	σ_y
	Shareholder		
(6)	SOP failure rate	$\mathbb{E}[\text{SOP fail}]$	χ_s
(7)	SOP fail wage sensitivity	s_1	$\chi_s,\ \chi_b$
(8)	SOP fail output shock sensitivity	s_2	σ_{z_s}
(9)	Excess vote-against variance	$\mathrm{Var}\left(\tilde{V}^{\text{against}}\right)$	σ_{z_s}
	Board		
(10)	Average log wage	$\mathbb{E}[\log w]$	λ_0
(11)	Change in log wage (SOP fail)	b_1	χ_b
(12)	Wage growth over tenure	b_2	λ_1
(13)	Wage FE variance	$\mathrm{Var}\left(\mu^b\right)$	$\sigma_{z_b},\ c_w$
(14)	Log wage autocovariance	$\mathrm{Cov}(\log w, \log w_{-})$	c_w
(15)	Wage FE variance tenure slope	v_1	σ_{z_b}

Panel B. Pairwise correlations between empirical moments

		(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)
$\mathbb{E}[\log y]$	(1)	1.00														
k_1	(2)	-0.08	1.00													
y_1	(3)	-0.09	0.24	1.00												
$\mathrm{Var}(\mu^y)$	(4)	0.19	0.15	0.03	1.00											
$\mathrm{Var}(\epsilon^y)$	(5)	0.06	0.18	0.02	0.72	1.00										
$\mathbb{E}[\text{SOP fail}]$	(6)	0.00	-0.09	-0.10	0.06	0.17	1.00									
s_1	(7)	-0.05	-0.08	0.06	-0.09	0.02	0.42	1.00								
s_2	(8)	-0.05	-0.06	0.10	-0.09	0.01	0.02	0.50	1.00							
$\mathbb{E}[\log w]$	(9)	0.76	0.03	0.01	0.10	0.07	0.17	0.13	0.01	1.00						
b_1	(10)	-0.08	-0.05	-0.17	-0.07	0.04	0.05	0.40	0.13	-0.03	1.00					
b_2	(11)	-0.13	0.11	0.46	0.01	0.02	-0.07	0.02	0.04	-0.08	0.06	1.00				
$\mathrm{Var}\left(\mu^b\right)$	(12)	-0.06	-0.06	-0.03	0.24	0.02	0.00	-0.09	-0.03	-0.28	-0.03	-0.05	1.00			
$\mathrm{Cov}(\log w, \log w_{-})$	(13)	-0.01	-0.02	0.04	0.23	0.06	0.03	-0.09	-0.04	-0.23	-0.04	0.02	0.77	1.00		
$\mathrm{Var}\left(\tilde{V}^{\text{against}}\right)$	(14)	-0.16	0.10	0.13	-0.13	-0.12	-0.72	-0.27	0.01	-0.13	0.00	0.11	-0.12	-0.13	1.00	
v_1	(15)	0.21	-0.00	-0.04	0.04	-0.04	-0.01	-0.01	0.02	0.21	-0.01	-0.08	0.06	0.19	-0.03	1.00

Table A.2. Parameter–moment sensitivity for key parameters

This table reports the standardized sensitivity matrix of Andrews et al. (2017),

$$\tilde{\Lambda}_{kj} = -\left[(G'WG)^{-1}G'W\right]_{kj}\,\hat{\sigma}(m_j)/\hat{\sigma}(\hat{\theta}_k),$$

evaluated at the estimated parameter vector $\hat{\theta}$, where G is the moment Jacobian and W is the SMM weight matrix. Each row describes how the joint SMM estimator of parameter k responds to a one–standard-deviation perturbation in each moment, with all other parameters in the focus block and all other moments treated symmetrically through the joint estimation. Standardized values lie in $[-1, 1]$, with larger absolute magnitudes indicating that moment j contributes more local information to $\hat{\theta}_k$ in the joint problem. To focus on the parameters most relevant to the SOP environment, the table is restricted to six parameters ($\chi, \chi_s, \sigma_{z_b}, \sigma_{z_s}, \lambda_0, \lambda_1$) and moments important for identification of those parameters. G is computed on the cluster by two-sided finite differences with relative step 10^{-2}, averaged across 100 simulation seeds; see Appendix IA2.3.

	σ_{z_s}	χ_s	σ_{z_b}	χ_b	λ_0	λ_1
$\mathbb{E}$[SOP fail]	0.11	0.54	0.12	0.06	-0.09	-0.08
s_1	-0.14	-0.54	-0.08	0.10	0.22	-0.20
s_2	-0.38	0.28	0.30	-0.10	-0.16	0.16
$\mathrm{Var}\left(\tilde{V}^{\mathrm{against}}\right)$	0.14	-0.02	-0.09	0.12	0.14	-0.13
$\mathbb{E}[\log w]$	0.24	-0.18	-0.60	0.02	0.81	0.21
b_1	0.82	0.23	-0.35	-0.33	-0.09	-0.36
b_2	0.16	0.28	0.08	0.08	-0.14	0.35
$\mathrm{Var}\left(\mu^b\right)$	-0.20	-0.41	0.61	0.02	-0.26	0.59
v_1	0.05	-0.09	-0.17	0.92	-0.39	0.52

B. Counterfactual Appendix

This section outlines the technical details of the information-pass-through counterfactual in Section 6. The counterfactual nests the full-information revise-and-resubmit regime and the pure advisory implementation rule using a communication-friction parameter $\rho \in [0, 1]$. When $\rho = 0$, failure fully implements the information-pooled wage w_t^{info}. When $\rho = 1$, failure has no within-period effect on pay, as in the advisory regime. Intermediate values of ρ partially incorporate vote-revealed shareholder information into the revised wage.

Timing and information. The counterfactual modifies the within-period timing of the vote and, as a result, the information that enters the voting decision. In the baseline advisory regime (Section 3.2), the Board sets pay before output is realized, and the Say-on-Pay vote occurs at the annual meeting after output and productivity are revealed; thus the vote can condition on the public productivity signal $z_{yt} = \ln A_t$ from (11) in addition to shareholders' private signal z_{st} from (10). This is reflected in the baseline vote signal aggregation in (15) and the induced failure probability in (17).

In the counterfactual, the vote is moved forward to occur immediately after the compensation committee meeting and before operations take place. Consequently, when voting, shareholders do not observe output and do not observe the productivity signal z_{yt} from (11). The vote therefore conditions on the proposed wage and shareholders' private signal z_{st}, together with any information inferred from the proposal, but not on within-period performance realizations. This timing change is the main informational difference between the counterfactual and the baseline.

Let w_t^{info} denote the wage the Board would set if it observed the shareholder signal z_{st} in addition to its own signal z_{bt}, while retaining the same pay-setting objective as in the baseline. In particular, the governance wedge λ_τ remains active in the revision decision. The counterfactual therefore routes shareholder information to a still-biased Board; it does not impose the shareholder-optimal wage. The communication-friction parameter ρ governs how much of this information-pooled wage is incorporated after a failed vote. I define the failure-state wage as

$$\log w_t^{\text{fail}}(\rho) = \rho \log w_t^{\text{pass}} + (1-\rho)\log w_t^{\text{info}}, \qquad \rho \in [0,1]. \tag{B.1}$$

Thus, ρ measures the friction in passing vote-revealed information into pay-setting. At $\rho = 0$, $w_t^{\text{fail}}(\rho) = w_t^{\text{info}}$, so failure fully implements the information-pooled wage chosen by the Board under its baseline objective. At $\rho = 1$, $w_t^{\text{fail}}(\rho) = w_t^{\text{pass}}$, so failure does not change within-period pay and the regime is advisory in implementation. Intermediate values partially adjust the failure-state wage toward w_t^{info}. The gains from this exercise should therefore be interpreted as the value of improving information transmission to the Board, not as the value of removing the Board–shareholder preference wedge.

The Shareholder's problem. As in the baseline, shareholders commit ex ante to a voting policy indexed by s_t (Section 3.3), taking expectations over within-period uncertainty and integrating over the Board's private signal z_{bt}, which is realized before the wage proposal. The key difference from the baseline shareholder problem in (18) is that vote failure may now change implementation within the period. Shareholders therefore internalize a pass–fail payoff difference, net of the baseline cost of dissent.

Let $\tilde{F}_t^{S}(s_t \times \mathrm{w})$ denote the probability of failure induced by the threshold rule, defined analogously to (17) but with the counterfactual information set at the vote: because the vote occurs before A_t and y_t are realized, $\tilde{F}_t^{S}(\cdot)$ is induced by shareholders' private signal z_{st} and the proposal, rather than by the combined signal (z_{st}, z_{yt}) in (15). Define the firm's operating income under shareholders' information as

$$\tilde{\pi}_{st}(\mathrm{w}) = \mathbb{E}_{st}[A_t]\mathrm{w}^{\alpha} - \kappa\,\mathrm{w},$$

where the expectation is taken conditional on shareholders' information at the voting stage, which excludes the realized productivity signal z_{yt} in (11).

Given the Board's proposed wage w_t^{pass}, shareholders choose s_t to maximize expected operating income under the wage implemented in each vote outcome:

$$\max_{s_t} \quad \left(1 - \tilde{F}_t^{\mathrm{S}}\left(s_t \times w_t^{\mathrm{pass}}\right)\right) \times \underbrace{\int_{z_b} f(z_b)\left[\tilde{\pi}_{st}\left(w_t^{\mathrm{pass}}\right)\right] dz_b}_{\text{Pass: CEO receives proposed wage}} + \\ \tilde{F}_t^{\mathrm{S}}\left(s_t \times w_t^{\mathrm{pass}}\right) \times \underbrace{\left(\int_{z_b} f(z_b)\left[\tilde{\pi}_{st}\left(w_t^{\mathrm{fail}}(\rho)\right)\right] dz_b - \chi_s\right)}_{\text{Fail: CEO receives } w_t^{\mathrm{fail}}(\rho) \text{ \& S pays cost } \chi_s}. \tag{B.2}$$

The chosen wage w_t^{pass} is a function of (z_{bt}, s_t, ρ), which is suppressed for notational convenience. Relative to (18), the term $\chi_s \times \tilde{F}_t^{\mathrm{S}}(\cdot)$ is no longer the only consequence of failure when $\rho < 1$: failure also changes the implemented wage from w_t^{pass} to $w_t^{\mathrm{fail}}(\rho)$, which changes operating income directly. When $\rho = 1$, this implementation effect disappears and failure matters only through the cost of dissent, as in an advisory vote.

The Board's problem. In the baseline, the Board chooses the implemented wage and trades off operating income, bias, adjustment costs, and the expected utility cost of SOP failure (see (20)). Under the counterfactual, the Board instead proposes w_t^{pass} anticipating that the implemented wage equals w_t^{pass} if the vote passes and equals $w_t^{\mathrm{fail}}(\rho)$ if the vote fails. Because failure may change implementation, the Board's objective must account for pay and continuation values in both branches.

Formally, under the counterfactual the Board proposes w_t^{pass} according to

$$\tilde{V}_\rho(\mu_{at}, \tau_t, w_{t-1}) = \max_{w_t^{\mathrm{pass}}}\left(1 - \tilde{F}_t^{\mathrm{S}}\left(s_t \times w_t^{\mathrm{pass}}\right)\right)\left[\tilde{\pi}^{\mathrm{B}}_{at|z_{bt}}(w_t^{\mathrm{pass}}) - AC(w_t^{\mathrm{pass}}, w_{t-1}; \tau_t) + \tilde{V}_{\rho,at|z_{bt}}(w_t^{\mathrm{pass}})\right] + \\ \tilde{F}_t^{\mathrm{S}}\left(s_t \times w_t^{\mathrm{pass}}\right)\left[\tilde{\pi}^{\mathrm{B}}_{at|z_{bt}}\left(w_t^{\mathrm{fail}}(\rho)\right) - AC\left(w_t^{\mathrm{fail}}(\rho), w_{t-1}; \tau_t\right) + \tilde{V}_{\rho,at|z_{bt}}\left(w_t^{\mathrm{fail}}(\rho)\right) - \chi_b\right], \tag{B.3}$$

where

$$\tilde{\pi}^{\mathrm{B}}_{at|z_{bt}}(\mathrm{w}) = \mathbb{E}_{at|z_{bt}}[A_t]\mathrm{w}^{\alpha} - \kappa\mathrm{w} + \lambda_\tau \kappa \mathrm{w},$$
$$\tilde{V}_{\rho,at|z_{bt}}(\mathrm{w}) = \mathbb{E}_{at|z_{bt}}\left[(1 - f_{\tau_t})V_\rho(\mu_{at+1}, \tau + 1, \mathrm{w}) + f_{\tau_t} V^R\right].$$

When $\rho = 0$, the failure branch evaluates pay and continuation values at the information-pooled wage w_t^{info}. When $\rho = 1$, the failure branch evaluates pay and continuation values at the same wage as the pass branch, so failure affects the Board only through the utility cost χ_b.

Solution. I solve the counterfactual by simulation, holding primitives fixed and using the same sequence of shocks as in the baseline. First, I solve for the information-pooled wage w_t^{info}: the wage the Board would set if it observed the shareholder signal z_{st} in addition to its own signal z_{bt}. Second, for each value of ρ, I construct the failure-state wage $w_t^{\text{fail}}(\rho)$ using (B.1). Third, I compute the Board's optimal proposal w_t^{pass} from (B.3), taking into account that the vote occurs before z_{yt} is realized and that failure implements $w_t^{\text{fail}}(\rho)$.

Internet Appendix

IA1. Model Appendix

IA1.1. Microfoundation of the production function and Board wage-setting

This appendix microfounds two reduced-form objects used in the main model: the pay–output technology

$$y = \eta A w^{\alpha}$$

and the Board–shareholder preference wedge λ_{τ}.

CEO input choice and the pay–output reduced form. The Board offers a linear contract that pays ω per unit of CEO input n, so total pay is

$$w \equiv \omega n.$$

The CEO chooses n to maximize

$$u(\omega, n) = \omega n - n^{1/\gamma}, \qquad \gamma \in (0, 1),$$

as in standard convex-effort-cost formulations (Edmans and Gabaix, 2016). The first-order condition is

$$\omega = \frac{1}{\gamma} n^{1/\gamma - 1},$$

which implies

$$n(\omega) = (\gamma\omega)^{\gamma/(1-\gamma)}.$$

Since $w = \omega n(\omega)$, CEO input can be written as a power function of total pay:

$$n(w) = \gamma^{\gamma} w^{\gamma}.$$

Output is produced according to

$$y(A, n) = \eta A n^{\beta}, \qquad \beta \in (0, 1),$$

where β captures diminishing returns to CEO input. Substituting $n(w)$ gives

$$y = \eta A (\gamma^{\gamma} w^{\gamma})^{\beta}.$$

Absorbing the constant $\gamma^{\gamma\beta}$ into η yields the reduced-form production function

$$y = \eta A w^{\alpha}, \qquad \alpha \equiv \beta\gamma \in (0, 1).$$

Thus, total compensation is a sufficient statistic for the CEO input induced by the compensation contract. Only the composite curvature $\alpha = \beta\gamma$ enters the reduced form, so β and γ are not separately identified.

Shareholder-optimal wage. Operating profits are

$$\pi(A, w) = \eta A w^{\alpha} - \kappa w.$$

The shareholder-optimal wage solves

$$w_S(A) = \arg\max_{w>0} \{\eta A w^\alpha - \kappa w\},$$

so

$$w_S(A) = \left(\frac{\alpha\eta A}{\kappa}\right)^{\frac{1}{1-\alpha}}.$$

Preference wedge and over-production. The Board does not maximize shareholder profits. A Board–shareholder preference wedge $\lambda_\tau \in [0, 1)$ lowers the Board's effective marginal cost of pay, so the Board solves

$$\max_{w>0} \{\eta A w^\alpha - \kappa(1 - \lambda_\tau)w\}.$$

The Board's chosen wage is

$$w_B(A, \lambda_\tau) = \left(\frac{\alpha\eta A}{\kappa(1 - \lambda_\tau)}\right)^{\frac{1}{1-\alpha}} = (1 - \lambda_\tau)^{-\frac{1}{1-\alpha}} w_S(A).$$

Hence, for any $\lambda_\tau > 0$,

$$w_B(A, \lambda_\tau) > w_S(A).$$

Because CEO input and output are strictly increasing in pay,

$$w_B > w_S \implies n_B > n_S \implies y_B > y_S.$$

However, w_S maximizes shareholder profits, so

$$\pi(A, w_B) < \pi(A, w_S).$$

The wedge therefore generates overpay and over-production relative to the shareholder-optimal benchmark: CEOs at more captured Boards are paid more and produce more output, but shareholders earn lower net profits. In the main model, λ_τ varies with CEO tenure τ.

Microfoundation: endogenous director selection. The wedge λ_τ can be interpreted through the endogenous director selection mechanism in Hermalin and Weisbach (1998). New directors are appointed over time, and the CEO may influence nominations through the nominating committee. As tenure increases, the CEO's bargaining position over marginal appointments strengthens, causing the board to become more CEO-friendly. This interpretation is consistent with Coles et al. (2014), who show that board co-option rises with CEO tenure.

By the managerial-power logic of Bebchuk and Fried (2003), a board with more CEO-friendly directors places greater weight on the CEO's payoff or faces lower effective costs of approving high compensation. In the reduced-form Board objective above, this maps to a positive wedge λ_τ. I parameterize the tenure profile as

$$\lambda_\tau = \lambda_0(1 + \lambda_1)^\tau,$$

with parameters restricted so that $\lambda_\tau \in [0, 1)$ over the observed tenure support. This functional form is a parsimonious parameterization of monotone tenure dependence rather than a structural derivation.

IA1.2. Representative shareholder with correlated information

This appendix justifies the reduced-form vote-failure probability,

$$\Pr(\text{fail} \mid w, \mathcal{I}) = F(\text{threshold}(w, \mathcal{I})),$$

used in the main model. The result is informational equivalence: with atomistic ownership and correlated shareholder signals, the realized vote share is a sufficient statistic for the common component of shareholder information, so the Board's optimization against a representative shareholder is informationally equivalent to optimizing against the realized vote. The shared component captures common research, stewardship norms, and proxy-advisor input.

The atomistic-and-correlated-signals environment is a tractability device, not a literal description of the SOP voting environment. In practice, top institutional holders and proxy advisors are pivotal at the 30% and 50% failure thresholds, and the strategic dimension of their voting decisions is studied directly in Barry et al. (2026). Here that strategic dimension is absorbed into the reduced-form $F(\cdot)$, allowing the analysis to focus on the disciplinary role of SOP: how the Board adjusts pay in anticipation of dissent, without taking a stand on the within-base coordination problem. With non-pivotal atomistic shareholders, the symmetric cutoff rule used below is a behavioral primitive rather than a derived best response.

Proposition IA.1 (Informational equivalence). *Consider a continuum of atomistic shareholders $i \in [0, 1]$ who each observe a signal*

$$z_i = \overline{z} + \varepsilon_i, \qquad \varepsilon_i \overset{iid}{\sim} N(0, \sigma_\varepsilon^2),$$

where the common component satisfies

$$\overline{z} = a + v, \qquad v \sim N(0, \sigma_{\overline{z}}^2).$$

Suppose shareholders use a symmetric cutoff rule: vote against *iff $z_i \leq k(w)$, where $k(w)$ is an increasing function of the proposed wage (e.g., $k(w) = s \times w$). Let p denote the fraction of votes cast against. Then, as the shareholder base becomes large,*

$$p = \Phi\left(\frac{k(w) - \overline{z}}{\sigma_\varepsilon}\right) \quad a.s.$$

Hence the vote share p is a sufficient statistic for the common signal $\overline{z}$; equivalently, observing p is informationally equivalent to observing the aggregate signal

$$z_S \equiv \overline{z} = k(w) - \sigma_\varepsilon \, \Phi^{-1}(p),$$

with $z_S = a + v$ and $v \sim N(0, \sigma_{\overline{z}}^2)$. Therefore the shareholder base can be modeled as a representative shareholder who observes z_S and applies the same cutoff rule.

Proof. Conditional on $\overline{z}$, each shareholder votes against with probability $q(\overline{z}) = \Pr(z_i \leq k(w) \mid \overline{z}) = \Phi((k(w) - \overline{z})/\sigma_\varepsilon)$. The $\{\varepsilon_i\}$ are i.i.d. conditional on $\overline{z}$, so a continuum law of large numbers gives $p \longrightarrow q(\overline{z})$. Since $q(\cdot)$ is strictly monotone in $\overline{z}$, $\overline{z}$ is recovered by inversion. See Barry et al. (2026) for a detailed proof. ■

Discussion. Proxy-advisor recommendations map naturally into this structure as shifts in the common component $\overline{z}$ or reductions in $\sigma_{\overline{z}}^2$. The inversion $z_S = k(w) - \sigma_\varepsilon \, \Phi^{-1}(p)$ is a post-vote statistical identity recovered from the realized vote share, not a decision rule: the Board sets w before observing p and uses only the reduced-form failure probability $F(\cdot)$ when forming expectations over vote outcomes. Gaussian signals and the symmetric cutoff are tractability choices; alternative signal structures that monotonically link the vote share to an aggregate signal would support the same reduced form.

IA1.3. Microfoundation of commitment to costly failed votes

The main model treats shareholder commitment to a costly voting policy as a primitive: at each meeting the shareholder plays a voting policy that the Board takes as given when choosing w. The institutional structure of US SOP voting turns this commitment into a self-enforcing equilibrium: *not* punishing overpay is observably costly, not only punishing is costly.

The mechanism rests on three institutional features. First, the effective shareholder side consists of large institutional investors (index funds, ETFs, pension funds, and large active mutual funds) for whom exit is mechanically constrained (index funds and ETFs are bound to hold constituent firms) or reputationally costly. Second, major institutions publish detailed pay-voting policies (e.g., BlackRock Investment Stewardship, Vanguard Investment Stewardship, CalPERS Global Governance Principles), and Form N-PX makes their realized votes publicly observable; deviating from a stated policy is reputationally costly through aggregate adherence across the institution's portfolio rather than through any single vote. Third, the 70% support threshold described in Section 2.1 is externally enforced: ISS treats outcomes below 70% as low-support, triggers heightened scrutiny of the compensation committee, and may recommend against directors at the subsequent meeting (ISS, 2022); institutional engagement and compensation-committee director turnover both shift sharply at this cutoff (Dey et al., 2024). SOP is the domain in which institutional dissent does occur (Iliev et al., 2015), even where index-fund engagement on other governance questions is attenuated.

This institutional story admits a relational-contract interpretation (Bull, 1987). Consider a stylized infinitely repeated interaction between the Board (B) and shareholders (S): in each period, B chooses either a disciplined wage w_P or an overpaying wage $w_F > w_P$. Under a grim-trigger strategy, S fails the vote at private cost X_S whenever $w = w_F$ and otherwise passes; if S does not fail after observing w_F, the disciplinary threat loses credibility and B pays w_F in all future periods. Failing is optimal for S whenever the contemporaneous cost is no larger than the discounted value of preserving the low-wage continuation:

$$X_S \le \delta\big(E[V_D] - E[V_{ND}]\big) = \frac{\delta}{1-\delta} E[w_F - w_P].$$

That is, costly vote failure is credible whenever the present value of sustained discipline exceeds the contemporaneous failure cost. The argument is illustrative rather than a derivation inside the main model, which has finite expected horizon through firm exit and CEO turnover and uses a continuous wage policy; the institutional story makes the primitive commitment empirically plausible.

Beyond credibility, shareholders prefer the SOP regime *ex ante* even though realized vote failure is costly. The disciplined equilibrium delivers a lower wage path at the cost of an expected vote-failure loss $\overline{p} \cdot X_S$, where $\overline{p}$ is the equilibrium failure probability. Letting ΔV denote the shareholder-value difference between the SOP and no-SOP regimes, SOP is ex-ante preferred whenever

$$\Delta V > \overline{p} \cdot X_S.$$

The inequality holds in the main-model equilibrium: $\overline{p}$ is small because the Board sets w to anticipate dissent (failure is costly but rare), and the wage discipline SOP induces is large enough to dominate the expected punishment loss. Failed votes are therefore valuable to shareholders ex ante even though they are costly ex post.

Section IA1.2 and this section address two complementary sub-problems of the SOP voting environment: the atomistic limit there isolates how the vote share aggregates a common signal, while the institutional structure here provides the commitment device. The main text uses the resulting reduced form: shareholders commit ex ante to a voting policy that pins down an anticipated probability of failure, which the Board internalizes when setting pay.

IA1.4. Evolution of Board and Shareholder Beliefs

I first detail two Propositions, which define how beliefs update in the model. Then I define exactly how Board and shareholder beliefs change within each period.

IA1.4.1. Evolution of Beliefs Period to Period

Prop. IA.2 shows how beliefs change from t to $t+1$. Prop. IA.3 describes the distribution of next period beliefs given today's beliefs, which is used when the Board calculates their (expected) continuation value. Figure IA.1 displays a detailed timeline that illustrates belief updating and the timing of strategies.

Proposition IA.2. *From period t to $t+1$, the variance of beliefs for both the Board and shareholders declines deterministically according to*

$$\sigma_a^2(\tau+1) = \left[\sigma_a^{-2}(\tau) + \sigma_{z_b}^{-2} + \sigma_{z_s}^{-2} + \sigma_y^{-2}\right]^{-1} \tag{IA.1}$$

where τ is the tenure of the CEO at year t. Equivalently, I can write the variance of beliefs about CEO ability entirely as a function of CEO tenure τ and model parameters

$$\sigma_a^2(\tau) = \sigma_0^2\left[1 + \tau\left(\kappa_{z_b}^{-1} + \kappa_{z_s}^{-1} + \kappa_y^{-1}\right)\right]^{-1} \tag{IA.2}$$

where $\kappa_{z_b} = \sigma_{z_b}^2/\sigma_0^2$, $\kappa_{z_s} = \sigma_{z_s}^2/\sigma_0^2$ and $\kappa_y = \sigma_y^2/\sigma_0^2$

Similarly, from period t to $t+1$, the mean of beliefs for both the Board and shareholders evolves according to

$$\mu_{at+1} = \sigma_a^2(\tau+1)\left[\frac{\mu_{at}}{\sigma_a^2(\tau)} + \frac{z_{bt}}{\sigma_{z_b}^2} + \frac{z_{st}}{\sigma_{z_s}^2} + \frac{z_{yt}}{\sigma_y^2}\right] \tag{IA.3}$$

Proof. The formulas are standard results in Bayesian learning (e.g., Pastor and Veronesi, 2009; Taylor, 2010).[1] The Board and shareholder reveal their signals each period. Thus, Board and shareholders share the same beliefs about the variance from period to period. ■

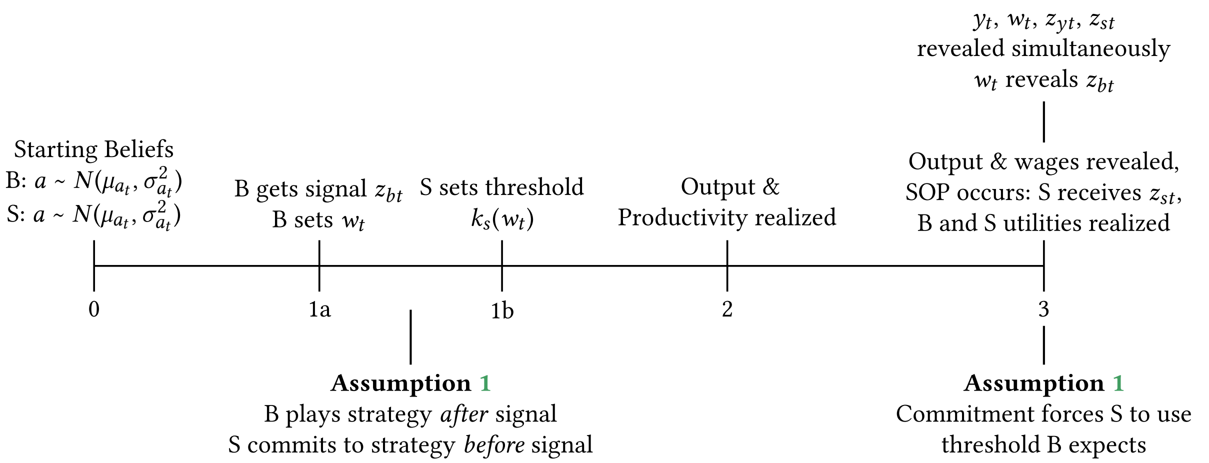
Proposition IA.3. *The mean and variance of the mean of $t+1$ CEO beliefs at t are*

$$\begin{aligned} \mathbb{E}_t[\mu_{at+1}] &= \mu_{at} \\ Var_t[\mu_{at+1}] &= \sigma_a^2(\tau) - \sigma_a^2(\tau+1) \end{aligned} \tag{IA.4}$$

[1] See also the internet appendix for Taylor (2010).

Figure IA.1. Model timeline with exact time that strategies are played

This figure displays a more detailed model timeline, with mappings to the relevant assumptions, and when the Board and Shareholder play their strategies. See Figure 1 in the main text for the timeline as it maps to real-world outcomes. See Appendix IA1.5.



That is,

$$\mu_{at+1} \mid \mu_{at}, \tau \sim N\left(\mu_{at}, \sigma_a^2(\tau) - \sigma_a^2(\tau+1)\right)$$

Proof. Again, this is a standard result in Bayesian learning, i.e., $\text{Var}(E[a \mid I'] \mid I) = \text{Var}(a \mid I) - \text{Var}(a \mid I')$, where a is a Normal process, and I and I' are information sets before and after realizations of normal signals about a. ■

IA1.4.2. Differences in Board and Shareholder Beliefs Within Period

This section describes how Board and shareholder beliefs evolve within each period. As the wage and vote reveal signals z_b and z_s, B and S share the same beliefs at the beginning of t. By Prop. IA.2, I have that $\sigma^2_{bt} = \sigma^2_{st} = \sigma^2_a(\tau_t)$ from (IA.2). Let μ_{at} be the beliefs about the mean at the beginning of period t. The evolution of B and S beliefs within t is:

1. **Board beliefs after the compensation committee meeting**
   At the meeting, the Board receives signal z_{bt}, and Board beliefs update to

$$\mu_{bt|z_b} = \sigma^2_{bt|z_b}\left(\frac{\mu_{at}}{\sigma^2_a(\tau_t)} + \frac{z_{bt}}{\sigma^2_{z_b}}\right) \tag{IA.5}$$

$$\sigma^2_{bt|z_b} = \frac{\sigma^2_a(\tau_t)\sigma^2_{z_b}}{\sigma^2_a(\tau_t) + \sigma^2_{z_b}} = \sigma^2_0\left[1 + (\tau_t+1)\kappa^{-1}_{z_b} + \tau_t(\kappa^{-1}_{z_s} + \kappa^{-1}_y)\right]^{-1} \tag{IA.6}$$

   The Board makes their wage decision based upon these beliefs.

2. **Shareholder beliefs when they commit to signal threshold k_{st}**
   At the time that S commits to voting strategy, B has beliefs $\left(\mu_{bt|z_b}, \sigma_{bt|z_b}\right)$ and S has $\left(\mu_{at}, \sigma^2_a(\tau_t)\right)$ (the prior). S can discern $\left(\mu_{bt|z_b}, \sigma_{bt|z_b}\right)$ for any z_{bt}, which they integrate out in the objective function (18).

3. **Board and shareholder beliefs about shareholders' *ex ante* signal distribution at the time of the SOP vote**

Before the SOP vote, when the shareholders commit to their threshold, both B and S know that the shareholders' aggregated signal will be

$$Z_{syt} = a + p\varepsilon_{st} + (1-p)\varepsilon_{yt} \tag{IA.7}$$

with $Z_{syt} \sim N\left(\mu_{at}, \sigma_{at}^2 + \frac{\sigma_{z_s}^2 \sigma_y^2}{\sigma_{z_s}^2 + \sigma_y^2}\right)$.[2] Notice that shareholder beliefs about a do not update to $N\left(\mu_{bt|z_b}, \sigma_{bt|z_b}^2\right)$ when computing the benchmark wage w_t^S. Although shareholders observe the proposed wage before voting, I treat the wage as the policy being evaluated rather than as an additional signal from which shareholders invert the Board's private information. This tractability assumption keeps the ex-ante voting-policy problem static: the shareholder benchmark is formed from meeting information, (z_{st}, z_{yt}), while the observed wage enters separately through the gap between the proposed wage and the benchmark. Shareholders nevertheless choose the voting policy s_t ex ante taking into account the Board's equilibrium wage response to its private signal.

4. **Board and shareholder beliefs after the annual shareholder meeting**
   The 10-K and compensation committee report reveals the wage to shareholders, hence reveals z_{bt}. Shareholders vote at the annual meeting and Z_{syt} and thus z_{st} are revealed. Hence, B and S beliefs update to $\left(\mu_{at+1}, \sigma_a^2(\tau_t + 1)\right)$, by Prop. IA.2.

IA1.5. Assumptions about Shareholder strategy

See Section 3.3 for discussion of the Shareholder's strategy. I model this threat as an *ex ante* probability that the SOP will fail, which is increasing in the wage, and I make the following assumptions:

Assumption 1. *Shareholders commit to their voting strategy in advance of the annual shareholder meeting.*

I assume that S commits to an *ex ante* probability of vote failure to influence the Board's wage decision, even though *ex post* failure is costly. Because of the information revelation structure of the model (Figure IA.1), S sets the probability of vote failure before seeing z_{bt}, z_{st}, or z_{yt}. The threat of vote failure does not need to be revealed to the Board before the annual shareholder meeting, however commitment forces S to play the threshold strategies that the Board expects. Unlike Kakhbod et al. (2023), there is no notion of cheap talk here. Commitment means S cannot choose an *ex ante* optimal non-zero failure probability and then renege at the shareholder meeting once the Board sets their wage.[3]

Assumption 2. *Shareholders are* myopic. *That is, the SOP vote is only influenced by today, and is not a fully dynamic problem.*

Effectively, this means that shareholders play a static game, while the Board plays a dynamic one. This assumption matches reality. There is ample evidence that voting in SOPs is influ-

[2]The variance of the signal is $Var(a + p\varepsilon_{st} + (1-p)\varepsilon_y) = \sigma_a^2 + p^2\sigma_{st}^2 + (1-p)^2\sigma_y^2$, where $p = \frac{\sigma_{z_s}^{-2}}{\sigma_{z_s}^{-2} + \sigma_y^{-2}}$. Expanding this expression out leads to the expression for the variance.

[3]Based on Assumption 1, Figure IA.1 provides a more detailed version of the model timeline (slightly adapting Figure 1). In particular, in period 1 (or 1a and 1b), B and S set their strategies. These strategies are not revealed at this time, but this timing convention defines the notion of the Board's informational advantage. In particular, the Board plays their strategy *after* receiving signal; the shareholder plays their strategy *before*. The assumption of commitment forces S to stick with the strategy that B expects.

enced by short-run outcomes, such as current firm or stock performance (see Fisch et al., 2018; Novick, 2019, 2020). This makes the solution method much simpler, as I only need to solve for one value function in the numerical solution.

IA1.6. Full Derivation of Model Solution

Proposition IA.4. *The Board's problem can be written as*

$$V(\mu_a, \tau, w_-) = \max_{w(z_b,s)} \quad \exp\left(\mu_{b|z_b} + 0.5\left(\sigma^2_{b|z_b} + \sigma^2_y\right)\right) w(z_b, s)^\alpha - (1-\lambda_\tau) w(z_b, s) - \chi_b F^U_{\tilde{z}_s}(s \times w(z_b, s)) - AC(w(z_b, s), w_-; \tau) + \delta_b \left[f(\tau_t) V^R + (1 - f(\tau_t)) \mathbb{E}_{b|z_b} \left[V(\mu'_a, \tau + 1, w(z_b, s)) \right] \right] \tag{IA.8}$$

where

- $\mu_{b|z_b}$ *and* $\sigma^2_{b|z_b}$ *are defined in* (IA.5) *and* (IA.6),
- $F^U_{\tilde{z}_s}(s \times w)$ *is the CDF of the shareholder-optimal wage distribution evaluated at the threshold* $s \times w$,
$$w^S \sim \log N\left(m^S_t, v^S_t\right),$$
where m^S_t *and* v^S_t *are defined in the main text in the derivation of the shareholder's vote decision.*
- $AC(w, w_-; \tau)$ *(adjustment cost) is defined in* (5),
- $f(\tau_t)$ *are CEO tenure-specific hazard rates, with* $f_0 = 0$ *and* $f(\tau_t) = 1$
- $V^R = V(\mu_0, 0, 0)$ *as in* (21)
- $\mu'_a \mid \mu_{b|z_b}, \tau \sim N\left(\mu_{b|z_b}, \sigma^2_{b|_{z_b}} - \sigma^2_a(\tau + 1)\right)$ *from Prop.* IA.3

The shareholder's problem can be written as

$$\max_s \int_{z_b} f(z_b \mid \mu_a, \sigma^2_a) \left[\exp\left(\mu_{b|z_b} + 0.5\left(\sigma^2_{b|z_b} + \sigma^2_y\right)\right) w(z_b, s)^\alpha - w(z_b, s) - \chi_s CDF^S_s(s \times w(z_b, s)) \right] dz_b \tag{IA.9}$$

where $f\left(z_b \mid \mu_a, \sigma^2_a\right)$ *is the density function of* z_b *given prior beliefs about CEO ability, and all other objects are defined as above.*

Proof. I start with (IA.8). $\mu_{b|z_b}$ and $\sigma^2_{b|z_b}$ are Board beliefs after receiving their signal, hence are known from the perspective of the Board. As $A = \exp(a + \varepsilon_y)$, with a (and beliefs about a) normally distributed, I can write its expectation in terms of means and variances. The probability of vote failure is described in Section 3.3, but as a brief overview it is given by the CDF $F^U_{\tilde{z}_s}$, of the unbiased (lognormal) wage of beliefs implied by realizations of $\tilde{z}_s$. The adjustment cost makes the Board's problem dynamic, as they have to factor in the effect of wages on the continuation value.

V^R is value if the CEO retires, so beliefs reset and there is no adjustment cost. In other words, the Board's problem reverts to its $t = 1$ value; it is constant for any state as the prior belief of ability about the CEO talent pool is distributed $N(\mu_0, \sigma^2_0)$ for any state.

The distribution of μ_a' conditional on μ_a and τ is given in Prop. IA.3. However, because the Board has beliefs $\left(\mu_{b|z_b}, \sigma^2_{b|z_b}\right)$, the variance of next period *mean* beliefs (not the variance of beliefs) at the time the Board makes their decision is $\sigma^2_{b|z_b} - \sigma^2_{a'}$. This quantity will be used to take expectation over the continuation value. If the CEO continues, the tenure increases by 1, and the Board must consider the adjustment cost in the next period.

For (IA.9), the objects are the same as the Board's problem, however the shareholders choose s after integrating out z_b. That is, shareholders figure out the Board's wage decision for each z_b, including how they would react to the choice of a particular s, and maximize expected operating income. Under Assumption 2, shareholders do not behave dynamically, and only vote on the current period.

The solution $(w(z_b), s)$ is to be found numerically, each $(w(z_b), s)$ is a best response under commitment (Assumption 1). To sketch the intuition of the solution, fix S' strategy s under commitment. The Board can then back out the probability of failure for each choice of $w(z_b)$, knowing that S must play the threshold. In other words, there is no notion of deviation for the Board. S just needs to maximize (IA.9) for their strategy to be a best response; they cannot deviate at the vote and play a lower threshold. ■

IA1.7. Derivation of Model Statistics

IA1.7.1. Average shareholder value

To interpret the magnitude of the failure-cost parameters, I normalize them by an average level of shareholder value. I define *average shareholder value* as the present value of expected operating profits generated by a CEO of average skill when compensation is chosen to maximize shareholder surplus (i.e., absent SOP considerations and any difference in preferences between the Board and shareholder).

Shareholder-optimal wage. Per-period operating profits in the model are

$$\Pi_t = y_t - \kappa\eta w_t, \qquad y_t = \eta A_t w_t^{\alpha},$$

so $\Pi_t = \eta(A_t w_t^{\alpha} - \kappa w_t)$. For a CEO of average skill, the shareholder chooses a constant wage $w^\star$ to maximize expected operating profits:

$$w^\star \in \arg\max_{w>0} \; E[A]\, w^{\alpha} - \kappa w.$$

The first-order condition implies

$$w^\star = \left(\frac{\alpha}{\kappa} E[A]\right)^{\frac{1}{1-\alpha}}. \tag{IA.10}$$

Average productivity. Let $\log A$ be normally distributed for a CEO of average skill. Denoting its mean by μ_0 (normalized to zero in estimation) and its variance by σ_A^2, we have

$$E[A] = \exp\!\left(\mu_0 + \tfrac{1}{2}\sigma_A^2\right). \tag{IA.11}$$

σ_A^2 aggregates the relevant sources of cross-sectional and transitory dispersion in productivity used in the wage/value benchmark (including output risk σ_y^2 and the dispersion implied by private information). As such, the volatility reflects a perceived ex ante average productivity, incorporating that both output and uncertainty are resolved after the Board and Shareholder act on noisy information about CEO ability.[4]

Average shareholder value. Under a stationary benchmark with constant expected operating profits, average shareholder value is the discounted value of profits:

$$\overline{V} \equiv E\left[\sum_{t=0}^{\infty} \delta_b^t \Pi_t\right] = \frac{1}{1-\delta_b} E[\Pi], \tag{IA.12}$$

where $\delta_b \in (0,1)$ is the discount factor. Evaluated at the shareholder-optimal wage $w^\star$, expected per-period operating profits are

$$E[\Pi] = \eta(E[A](w^\star)^\alpha - \kappa w^\star) = \eta(1-\alpha)\, E[A]\,(w^\star)^\alpha,$$

where the last equality uses the first-order condition. Substituting (IA.10)–(IA.11) into (IA.12) yields a closed-form expression for $\overline{V}$ as a function of $(\alpha, \kappa, \eta, \mu_0, \sigma_A^2, \delta_b)$.

Normalizing unscaled structural parameters. I report all unscaled structural parameters in units of average shareholder value. For any unscaled parameter x, I report the normalized quantity $x/\overline{V}$, interpreted as the fraction of average shareholder value associated with the corresponding parameter. In Table 3, standard errors for these normalized parameters are computed via the Delta method, using the estimated variance–covariance matrix of the underlying structural parameters, noting that $\overline{V}$ can be estimated in closed form solely as a function of model parameters.

IA2. Estimation Appendix

IA2.1. Numerical Solution

The model requires 12 estimable parameters, the Board's discount rate and $\mathbf{T}+1$ externally calibrated CEO separation rates. I externally calibrate the Board's discount factor $\delta_b = 0.9615$, which corresponds to annual data. The CEO separation rates are generated by calculating the cross-sectional proportion of CEOs that separate from their firm for a given tenure. I group the remaining 12 parameters as $\mathbb{P}$,

$$\mathbb{P} = \begin{bmatrix}\log\eta & \kappa & \sigma_0 & \sigma_y & \alpha & c_w & \sigma_{z_b} & \sigma_{z_s} & \lambda_0 & \lambda_1 & \chi_b & \chi_s\end{bmatrix} \tag{IA.1}$$

The model's solution proceeds as such

1. Start with a given $\mathbb{P}$
2. Discretize each idiosyncratic shock into an N_z grid; e.g., fix the possible realizations of $\varepsilon_{z_{bt}}, \varepsilon_{z_{st}}, \varepsilon_{z_{yt}}$, while maintaining the same seed sequence across iterations. I set $N_z = 21$
3. Discretize the state space into a $\left(N_\mu, \mathbf{T}+1, N_w\right) = (21, 26, 15)$ grid, call it $\mathcal{S}$, where each tuple (μ_i, τ_j, w_k) indexes current mean belief about CEO ability, tenure (which fully determines beliefs of variance of CEO ability) and the previous wage.
4. Start with a guess of $V_0(\mu, \tau, w)$ as the solution to the static game (i.e., where there is no wage adjustment cost), so V_0 is just the Board's per-period expected utility given optimal choices. Each $V(\mu, \tau, w_{1:N_w})$ starts with the same value.

[4]This corresponds to $E[A] = \exp\left(\frac{1}{2}(\sigma_{0,\text{priv}}^2 + \sigma_y^2)\right)$ with $\sigma_{0,\text{priv}}^2 = \sigma_0^2 + \sigma_{\text{priv}}^2$ and $\sigma_{\text{priv}}^2 = (\sigma_{zb}^2 \sigma_{zs}^2)/(\sigma_{zb}^2 + \sigma_{zs}^2)$.

5. Use Gauss quadrature, interpolation and Prop. IA.2 to estimate the continuation value for each tuple (μ, τ, w)
6. Guess S' voting policy and compute $\mathcal{S}(\mu, \tau, w_{-})$, or S' flow utility given the voting policy
7. Given S' policy, find B's wage policy $\omega(\mu, \tau, w_{-})$ given S' policy using value function iteration on $V(\mu, \tau, w_{-})$
8. Return to 5 and repeat until $\max|V_i - V_{i-1}| < \epsilon = 1\text{e}-5$ and $\max|\mathcal{S}_i - \mathcal{S}_{i-1}| < \epsilon = 1\text{e}-5$

This process returns the Board's wage policy for each element in the state space and each realization on the grid of ε_{z_b}. Concurrently, it returns the shareholder's policy for each element in the state space.

IA2.2. Simulation

I set $N_f = 2000$ firms. Given $a \sim (0, \sigma_0)$, I draw a CEO of skill a for each firm. A CEO spell is the length of time the CEO is matched with a firm. Each period, for each firm, I generate realizations of $\varepsilon_{z_{bt}}$, $\varepsilon_{z_{st}}$ and ε_{yt} (which are drawn under the same sequence of quasi-random numbers each simulation iteration). Given the state, I use the policies described in Section IA2.1 to generate optimal choices. Beliefs update given realizations of $\varepsilon_{z_{bt}}$, $\varepsilon_{z_{st}}$ and ε_{yt}. At the end of each period, for CEOs with tenure $\tau > 0$, they separate (via firing, quitting or retirement) with exogenous probability f_τ.

I generate $N_S = 5$ samples for each simulation. I "fix" randomness across different simulations. That is, each $n_s \in N_S$ sample has the same seed across iterations, only the variance of each CEO ability and each shock changes. Within each simulation n_s, I allow 20 years of burn-in data, and use 15 years as my analysis sample.

IA2.3. Estimation

I estimate the 12 parameters in (IA.1) As mentioned above, the Board's discount factor δ_b is calibrated to 0.9615, and separation rates are calibrated to match observed separation rates in the sample. I estimate the remaining model parameters by finding a vector $\mathbb{P}$ of parameters that minimizes the weighted distance between simulated and data moments. That is, given model moment $m(\mathbb{P})$ and data moments $m(X)$ and an appropriate weighting matrix W, I minimize

$$\min_{\mathbb{P}} \left[m(\mathbb{P}) - m(X)\right]' W \left[m(\mathbb{P}) - m(X)\right] \tag{IA.2}$$

Weight matrix. I use the inverse of the variance-covariance matrix of the data moments as the weight matrix in estimation: $W = V^{-1}$, which is the efficient weight matrix for SMM. To produce V, I use the influence function (IF) approach. I cluster V by firm-CEO spell, which is the appropriate clustering approach given that the model's learning and shock processes are all confined within CEO spell. Formally, for each moment m_k, let IF_{ik} be observation i's marginal effect on the estimated m_k. Then

$$\sqrt{N}(\hat{m} - m_0) = \frac{1}{\sqrt{N}} \sum_{i=1}^{N} IF_i + o_p(1)$$

and $\hat{V}$ can be consistently estimated via

$$\hat{V} = \frac{1}{N}\sum_{g=1}^{G}\left(\sum_{i\in g} IF_i\right)\left(\sum_{i\in g} IF_i\right)',$$

where g is a firm-CEO spell. The weight matrix used in the estimation is $\hat{W} = \hat{V}^{-1}$.

Moment Jacobian. The moment Jacobian $G = \partial m(\mathbb{P})/\partial\mathbb{P}'$ is computed by central differences, averaged across 100 simulation seeds. For each parameter, I take a two-sided finite difference with relative step 10^{-2}, holding the simulated panel size fixed at 5,000 firms and 50 simulation draws per evaluation. To reduce simulation noise, I repeat the calculation across 100 independent seeds and average the per-seed Jacobians; standard errors and the AGS sensitivity in Table A.2 use this averaged G.

IA2.4. Optimization algorithm

My goal is to find the global minimum of the SMM objective function described in Section IA2.3. To leverage the efficiency of parallel computing, I use a somewhat modified version of the TikTak global optimization algorithm described in Arnoud et al. (2019).[5] The modifications are designed to take advantage of high performance computing to minimize computing time. The global optimization routine can be described as such:

1. **Parallel local minimization**

   i. Generate bounds for each parameter. This is a holistic step, yet the bounds should be narrow enough to allow for the subsequent quasi-random sequences to adequately cover the space, but wide enough so that I maximize the chance of finding the global minimum.

   ii. Using the bounds, generate a Sobol sequence of length N. Sobol points are quasi-random points which are intended to mimic a draw from a uniform distribution. In my setup, I set $N = 50,000$ or $N = 100,000$.

   iii. For each $n \in N$ of the Sobol points, evaluate the objective function at n. Keep N_p of the points with the smallest initial function value. I set $N_p = 100$ or $N_p = 200$.

   iv. For each $p \in N_p$, solve for the local minimum. In my setup, I use Nelder-Mead locally, which gives N_p "promising" candidates for the global minimum.

2. **Parallel global minimization**. This step slightly modifies the TikTak routine to take advantage of parallel computing.

   i. Take the $p \in N_p$ candidates for the global minimum from above and sort in ascending order. Set $i = 1$, so the best minimum so far is indexed by i.

   ii. Take the best minimum so far, labeled p_i^*. Generate $N_p - i$ convex combinations using the TikTak methodology. That is, for $j \in N_p - i$, $p_{ji}^{cand} = \theta_{ji}p_i^* + (1-\theta_{ji})p_j$, where $\theta_{ji} \in [0,1]$ and approaches 1 as j increases.

   iii. Compute the local minimum of each $N_p - i$ point in parallel. If p_i^* is the best, then exit the routine and p_i^* is the candidate global minimum. Else,

[5] I modified code from https://github.com/tpapp/MultistartOptimization.jl, which is based upon the original TikTak code: https://github.com/serdarozkan/TikTak. See also Liu (2022) for a recent example.

iv. For the first j such that function value of p_{ji}^{cand} is less than that of p_i^*, stop all subsequent (unfinished) local minimization routines for $j' \in N_p - i$, and $j' > j$. Update i += p and return to ii.

This routine will return p_i^* as the global minimum.

3. **Polish global minimum**. Using stricter stopping criteria and a large number of function iterations, polish the global minimum p_i^* using a local minimization routine, e.g., Nelder-Mead.

References

Arnoud, A., Guvenen, F., and Kleineberg, T. (2019). Benchmarking global optimizers.

Barry, J. W., Pinnington, J., and Zhao, L. (2026). Information and preferences in shareholder voting.

Bebchuk, L. A. and Fried, J. M. (2003). Executive compensation as an agency problem. *Journal of economic perspectives*, 17(3):71–92.

Bull, C. (1987). The existence of self-enforcing implicit contracts. *The Quarterly journal of economics*, 102(1):147–159.

Coles, J. L., Daniel, N. D., and Naveen, L. (2014). Co-opted boards. *The Review of Financial Studies*, 27(6):1751–1796.

Dey, A., Starkweather, A., and White, J. T. (2024). Proxy advisory firms and corporate shareholder engagement. *The Review of Financial Studies*, 37(12):3877–3931.

Edmans, A. and Gabaix, X. (2016). Executive compensation: A modern primer. *Journal of Economic Literature*, 54(4):1232–1287.

Fisch, J., Palia, D., and Solomon, S. D. (2018). Is say on pay all about pay: The impact of firm performance. *Harvard Business Law Review*, 8:101.

Hermalin, B. E. and Weisbach, M. S. (1998). Endogenously chosen boards of directors and their monitoring of the CEO. *American Economic Review*, pages 96–118.

Iliev, P., Lins, K. V., Miller, D. P., and Roth, L. (2015). Shareholder voting and corporate governance around the world. *The Review of Financial Studies*, 28(8):2167–2202.

ISS (2022). United States proxy voting guidelines. https://www.issgovernance.com/file/policy/active/americas/US-Voting-Guidelines.pdf [Accessed: 2023-02-25].

Kakhbod, A., Loginova, U., Malenko, A., and Malenko, N. (2023). Advising the management: A theory of shareholder engagement. *The Review of Financial Studies*, 36(4):1319–1363.

Liu, T. (2022). Bargaining with private equity: Implications for hospital prices and patient welfare.

Novick, B. (2019). Executive compensation: The role of public company shareholders. https://corpgov.law.harvard.edu/2019/07/31/executive-compensation-the-role-of-public-company-shareholders/ [Accessed: 2023-02-18].

Novick, B. (2020). "The goldilocks dilemma": A response to lucian bebchuk and scott hirst. In *Colum. L. Rev. Forum*, volume 120, page 80. HeinOnline.

Pastor, L. and Veronesi, P. (2009). Learning in financial markets. *Annual Review of Financial Economics*, 1(1):361–381.

Taylor, L. A. (2010). Why are CEOs rarely fired? Evidence from structural estimation. *The Journal of Finance*, 65(6):2051–2087.